

# DRILLING FLUIDS *for* NON-MUD ENGINEERS

**MBW**  
FLUIDS



25 July 2026

*Protect GOOD Reservoir with GOOD MUD*

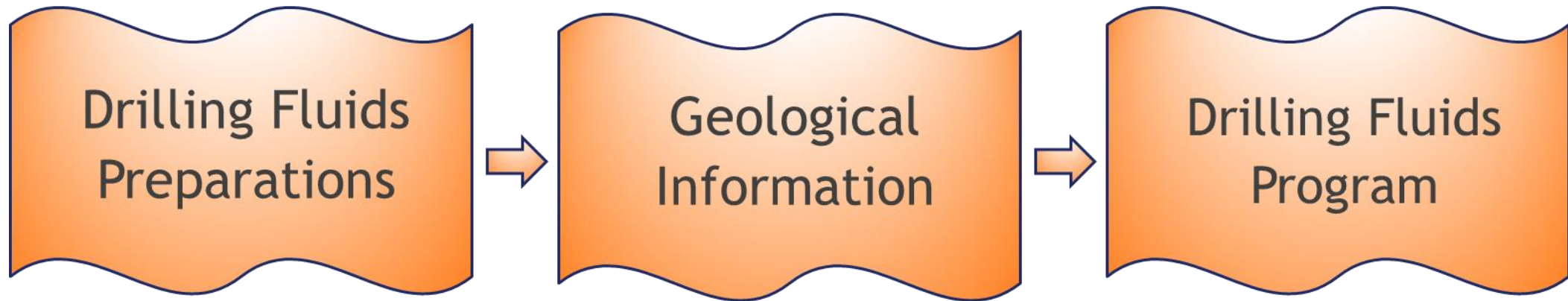
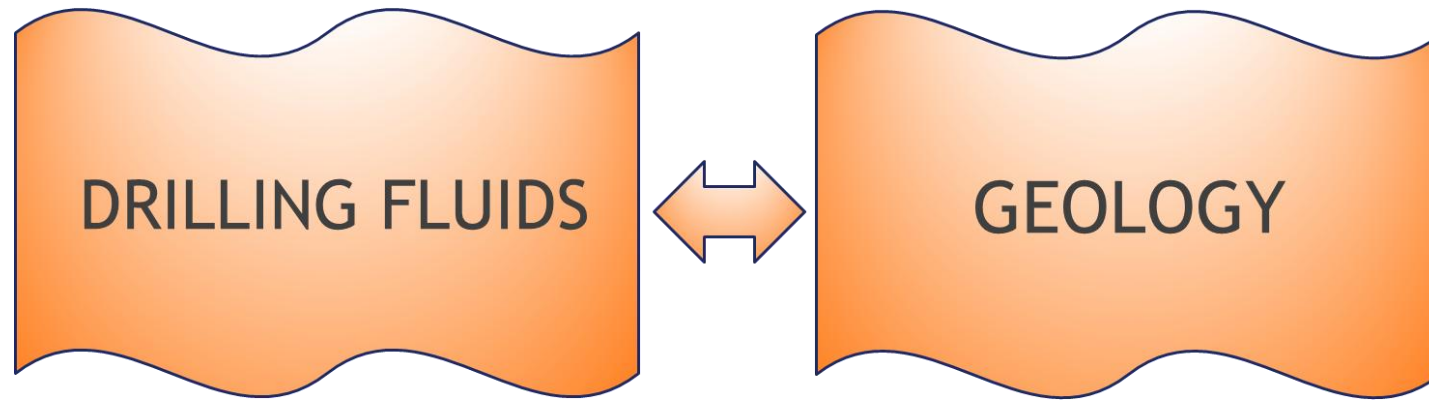
| Subjects |                                                                       |
|----------|-----------------------------------------------------------------------|
| 1        | Clay Mineralogy                                                       |
| 2        | Drilling Fluids:<br>Introduction, Functions, Classifications, Systems |
| 3        | Drilling Fluid Properties and Field Testing                           |



# Subject #1

## Clay Mineralogy





# The Objectives

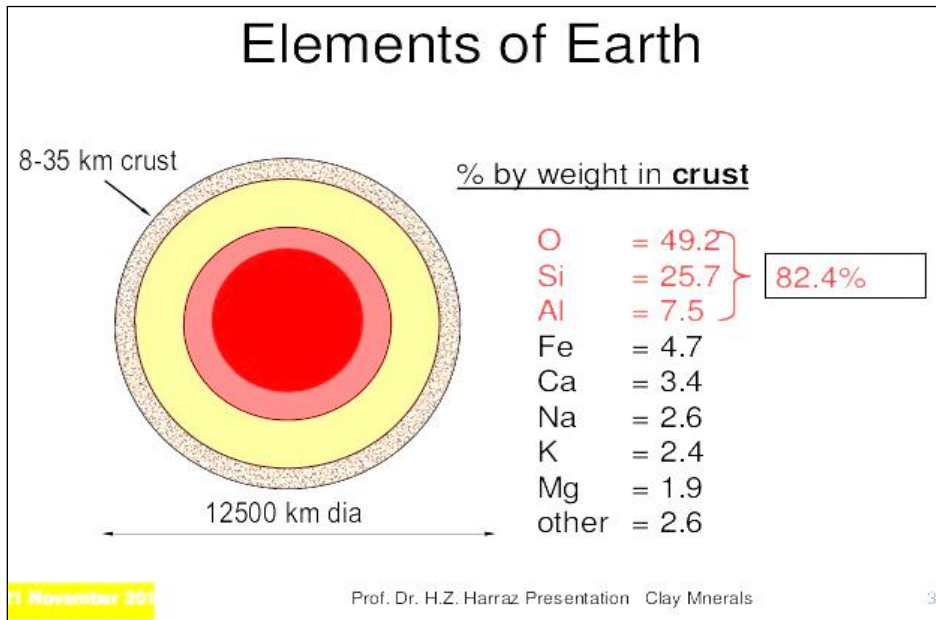
- \* A basic information to design a Drilling Fluid system
- \* To select correct products usage
- \* Foresee potential mud-related problems
- \* Approach to minimize formation damage
- \* Manage the cost-effective operations



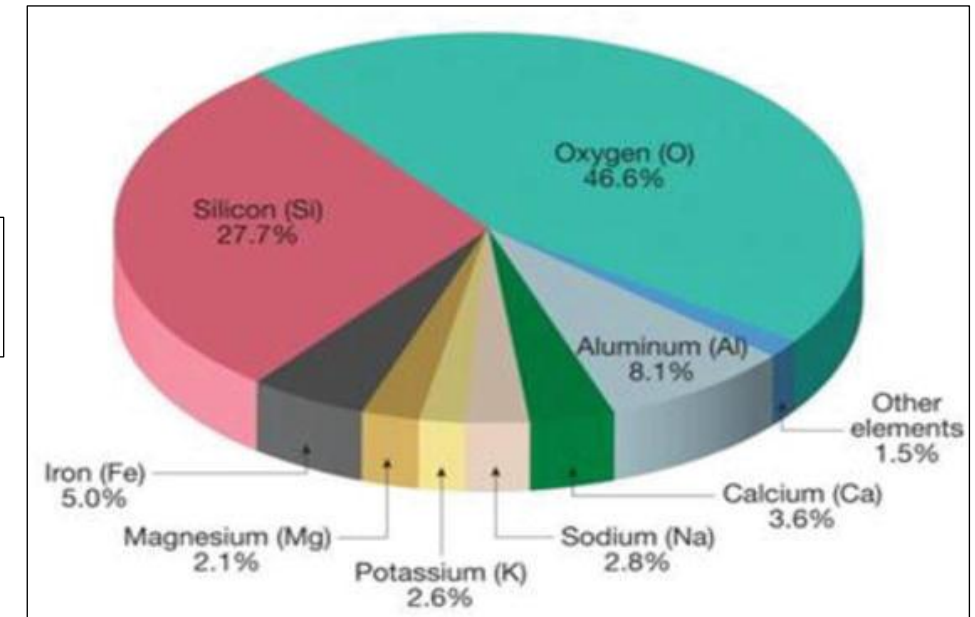
The most accurate “Drilling Fluid System”  
should be designed based on the  
“geological data” of the well to be drilled.



CLAY MINERALS: The most abundant sedimentary rocks, approximately 70-80% of the stratigraphic column. *Prof. Dr. Harraz, Clay Minerals, 15 Nov. 2015.*



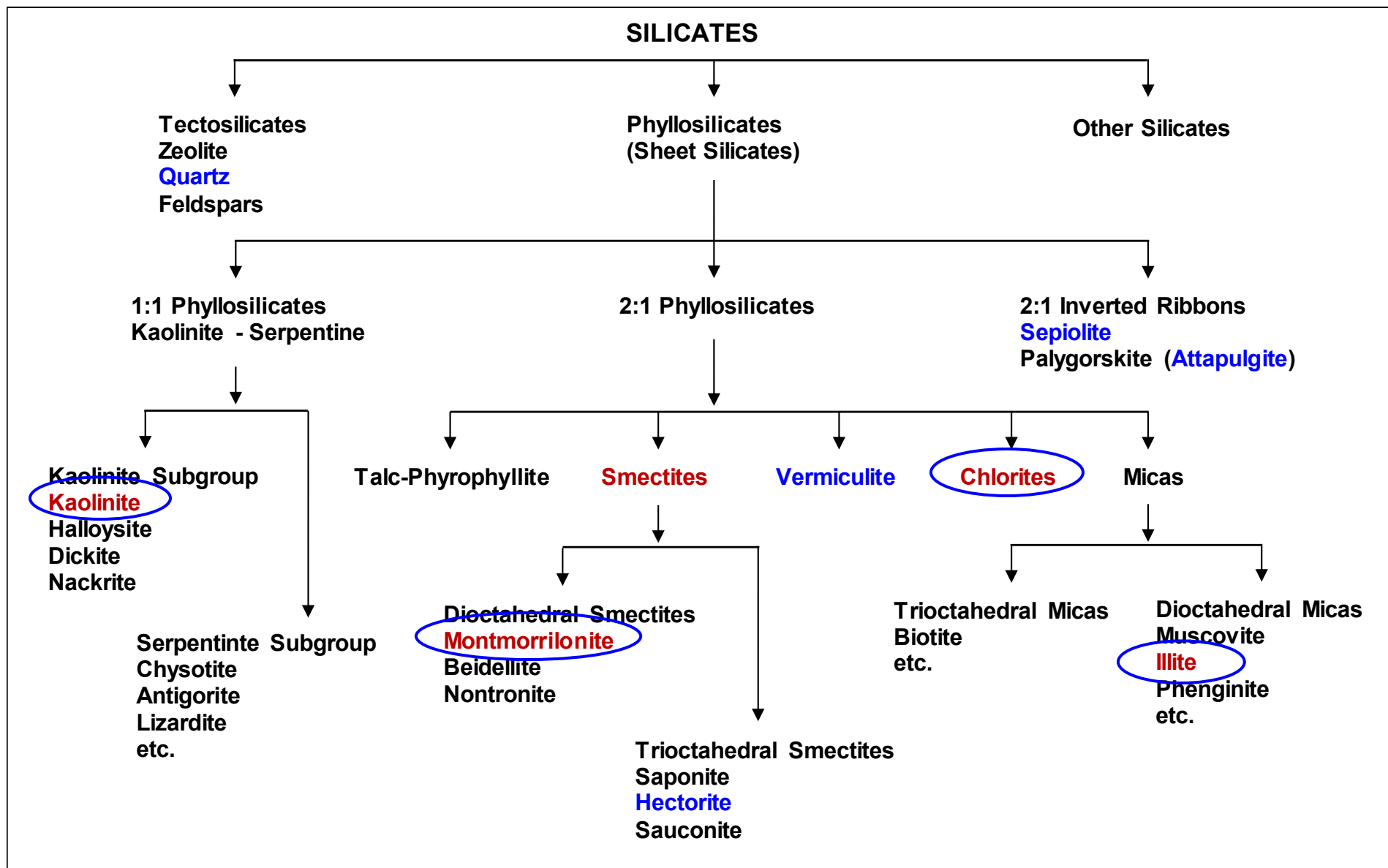
|           |   |       |
|-----------|---|-------|
| O = 46.6  | } | 82.4% |
| Si = 27.7 |   |       |
| Al = 8.1  |   |       |



# Clays and Clay Minerals

- \* “Clays” - commercial mined clays ~ BENTONITE:
  - \* *Sodium Bentonite (Wyoming Bentonite)*
  - \* *Calcium Bentonite*
- \* “Clay Minerals” - minerals present in sedimentary rocks:
  - \* Formation Clays:
    - \* *Kaolinite - Illite - Chlorite - Smectite (Montmorillonite)*
    - \* *Mixed-layer (Inter-layer): mixture of Illite/Smectite (I/S)*
    - \* Impacts: wellbore instability, stuck pipe, slow ROP, etc.





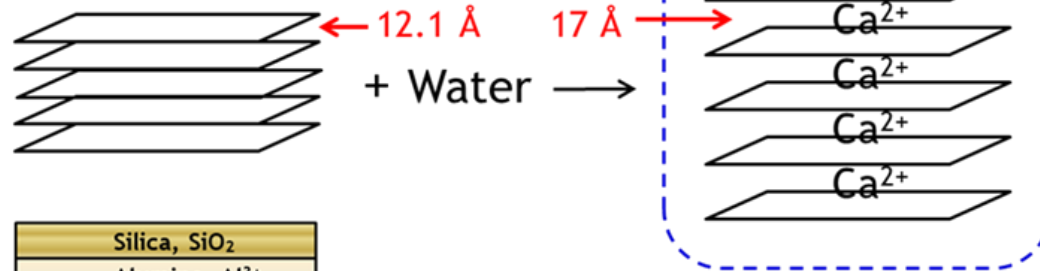
# Common Clays in Drilling Fluids Eng.

| Reactivity                                                                                                               | Clay Mineral                           | Descriptions                                                                                                                                                                                                                 | Drilling Fluid System<br>(if dominantly present)                                                                                                                                                                   |
|--------------------------------------------------------------------------------------------------------------------------|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Less</b></p>  <p><b>More</b></p> | <b>Kaolinite</b>                       | <ul style="list-style-type: none"> <li>* Particle: very hard, <b><i>non-swelling</i></b></li> <li>* In high percentage: brittle, easily break</li> <li>* Formed by weathering/hydrothermal alteration of Feldspar</li> </ul> | <ul style="list-style-type: none"> <li>* <b>Non-inhibited</b> mud system</li> <li>* Excessive salinity may break the clay into fine-particles.</li> <li>* Perform test to estimate salinity requirement</li> </ul> |
|                                                                                                                          | <b>Chlorite</b>                        | <ul style="list-style-type: none"> <li>* Particle: hard, <b><i>min-swelling</i></b></li> <li>* Tend to plug pore throats</li> </ul>                                                                                          |                                                                                                                                                                                                                    |
|                                                                                                                          | <b>Illite</b>                          | <ul style="list-style-type: none"> <li>* Particle: hard, <b><i>min-swelling</i></b></li> <li>* Potassium (K<sup>+</sup>) ions in interlayer capability.</li> <li>* Formed by weathering of Feldspar</li> </ul>               |                                                                                                                                                                                                                    |
|                                                                                                                          | Mixed layers:<br>Illite/Smectite (I/S) | <ul style="list-style-type: none"> <li>* Particle: <b><i>non-swelling</i></b></li> <li>* General composition: 70-80% Illite; 20-30% Smectite</li> </ul>                                                                      |                                                                                                                                                                                                                    |
|                                                                                                                          | <b>Smectite<br/>(Montmorillonite)</b>  | <ul style="list-style-type: none"> <li>* Particle: soft to very soft, <b><i>max-swelling</i></b></li> <li>* Water sensitive, highly reactive, dispersible</li> </ul>                                                         | <ul style="list-style-type: none"> <li>* <b>Inhibited WBM</b> system</li> <li>* <b>OBM</b> system</li> </ul>                                                                                                       |



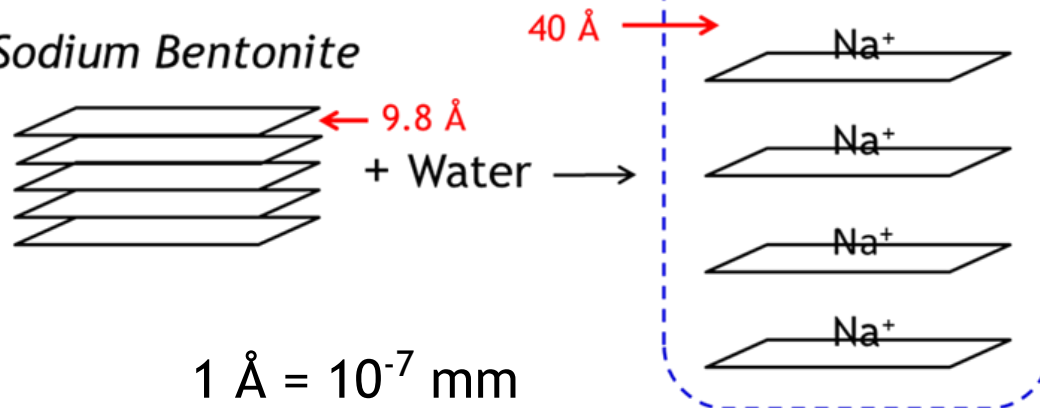
## Hydration Comparison of Calcium Bentonite and Sodium Bentonite

### Calcium Bentonite



|                           |
|---------------------------|
| Silica, SiO <sub>2</sub>  |
| Alumina, Al <sup>3+</sup> |
| Silica, SiO <sub>2</sub>  |

### Sodium Bentonite



$$1 \text{ Å} = 10^{-7} \text{ mm}$$

- \* **Calcium (Ca<sup>2+</sup>) Bentonite:**

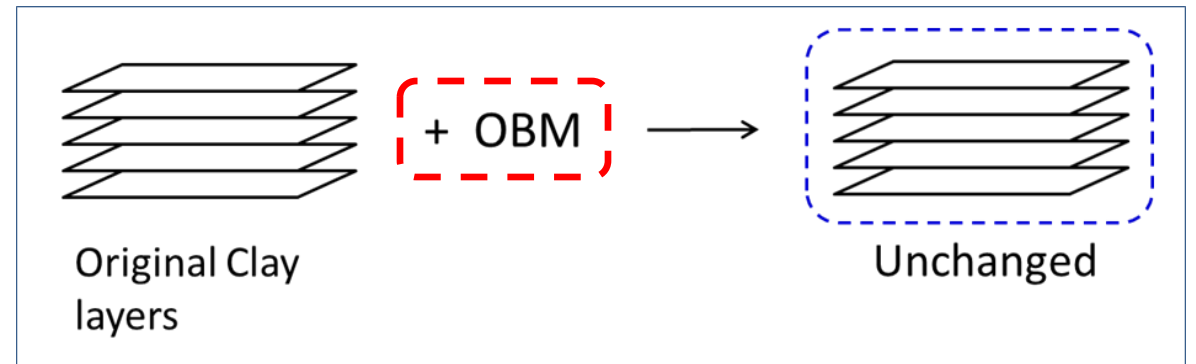
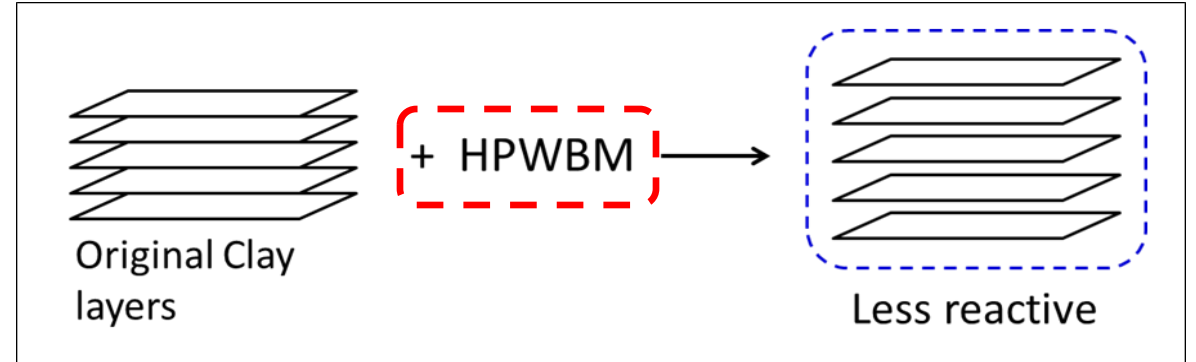
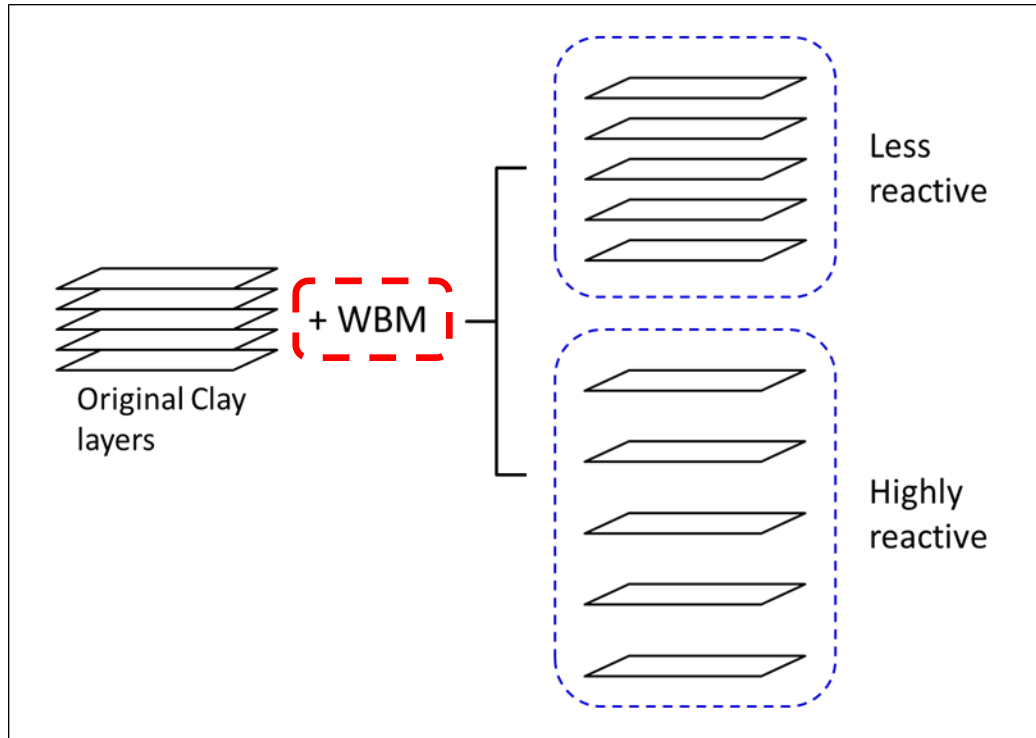
- \* Minimal expansion
- \* Less viscous

- \* **Sodium (Na<sup>+</sup>) Bentonite:**

- \* Maximal expansion
- \* High viscous



# Clays Reactivity in Different Mud Systems



# Shale

- \* Susceptibility to water
- \* Contains clays; troublesome shale contains high percentage of clays
- \* Expand with time
- \* Invisible to the naked eye
- \* Impacts: wellbore instability, stuck pipe, slow ROP, etc.



# Inhibition Tests



# Capillary Suction Timer (CST)

Purpose: estimate requirement of salinity for inhibition of a particular clay sample.



Fann Model 440



OFITE



# Linear Swell Meter (LSM)

- \* Measure the increase or decrease of an intact clay/shale core in “length over time”.
- \* The use of LSM and Capillary Suction Timer (CST) determine mud formulation for drilling through a specific clays/shale formation.



Fann Linear Swell Meter  
Model 2100



# Subject #2

## DRILLING FLUIDS

- \* An Introduction to Drilling Fluids
- \* Drilling Fluid Functions
- \* Drilling Fluid Classifications
- \* Drilling Fluid Systems



# Drilling Fluid: an Introduction



## *DRILLING FLUID by definition ....*

A mixture of “chemical additives” in a  
“liquid based”

**Chemicals + Liquid = DRILLING FLUIDS**



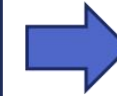
# Chemicals + Liquid = DRILLING FLUIDS

## CHEMICAL ADDITIVES: Powder + Liquid



## LIQUID BASED: Water or Oil

WATER Based:  
Fresh, Salty,  
Seawater



OIL Based:  
Diesel, Mineral  
Synthetic, Ester



# Drilling Fluid Functions



# Drilling Fluid Functions

1. Remove drilled cuttings from the wellbore
2. Stabilize pressures in the wellbore
3. Collect subsurface geological information
4. Allow interpretable electric logs
5. Minimize reservoir damage
6. Control corrosion
7. Support tubulars weight
8. Stabilize wellbore

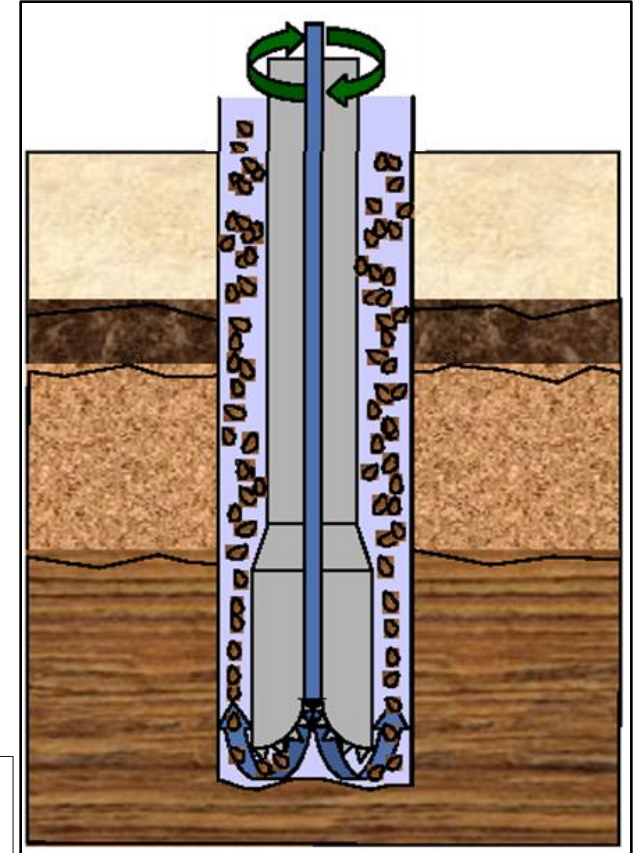


# Drilling Fluid Functions

1. Remove drilled cuttings from the wellbore



Drilled Cuttings



# Drilling Fluid Functions

## ② Stabilize pressures in the wellbore

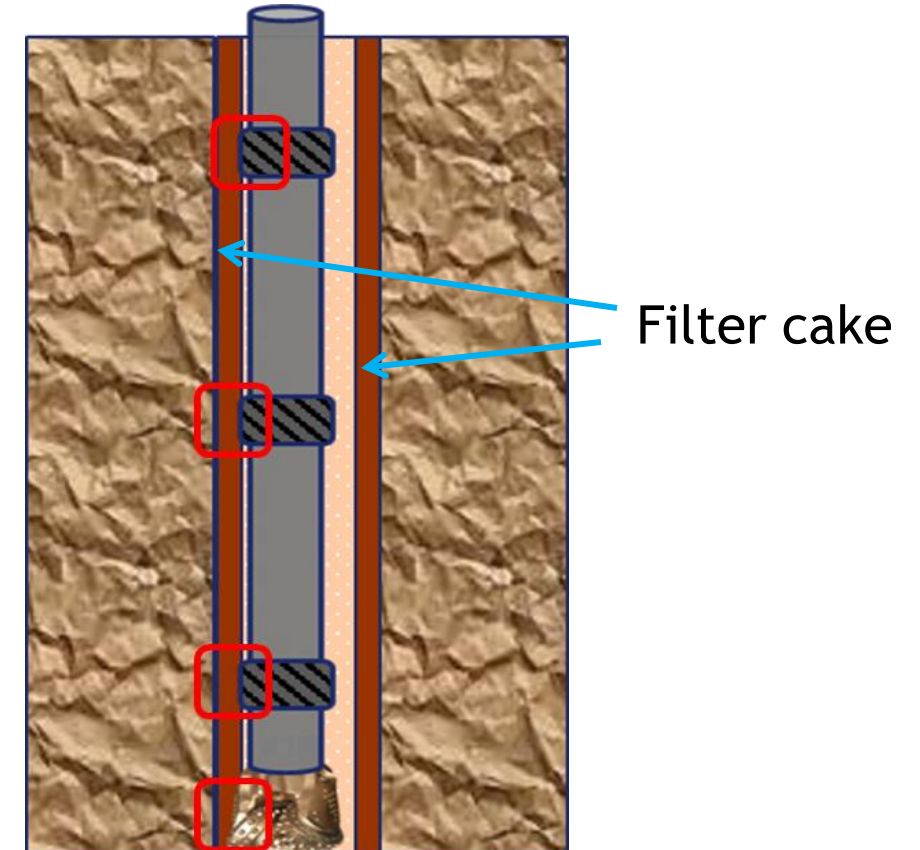
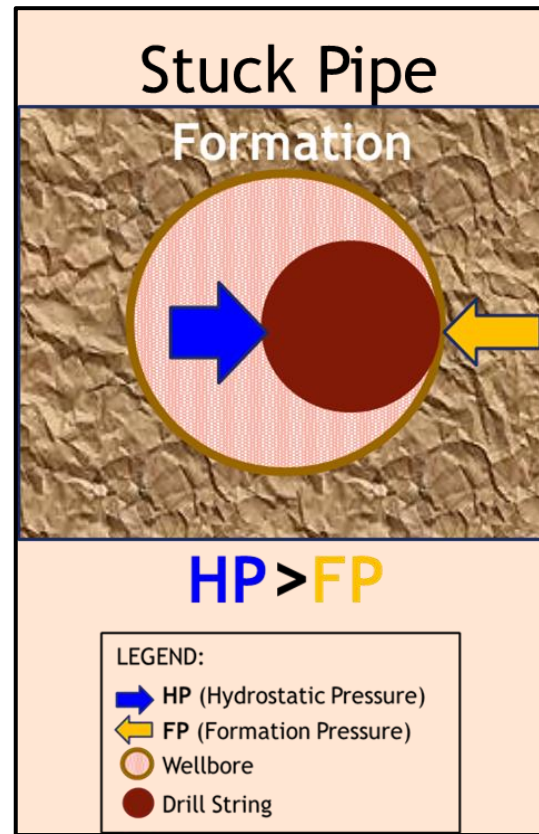
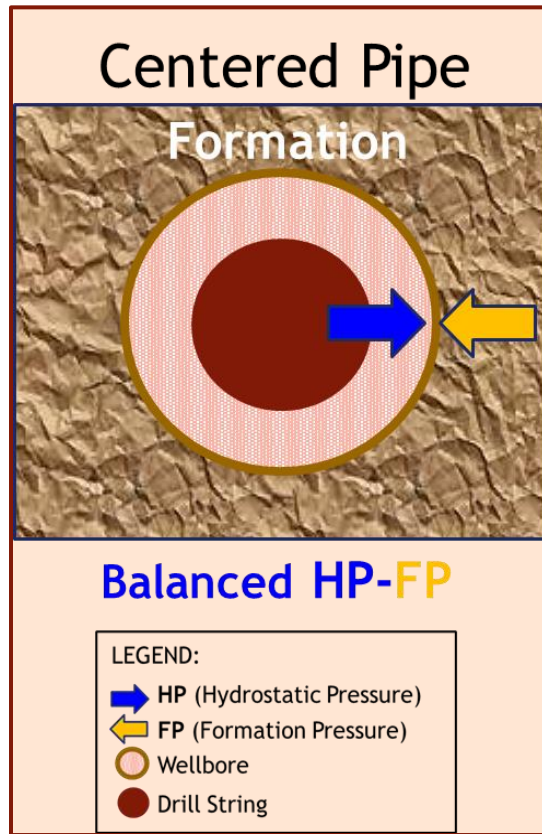
Impacts of “imbalance” pressures:

- \* Stuck pipe
- \* Loss of Circulation
- \* Gas Kick ~ Blow-Out



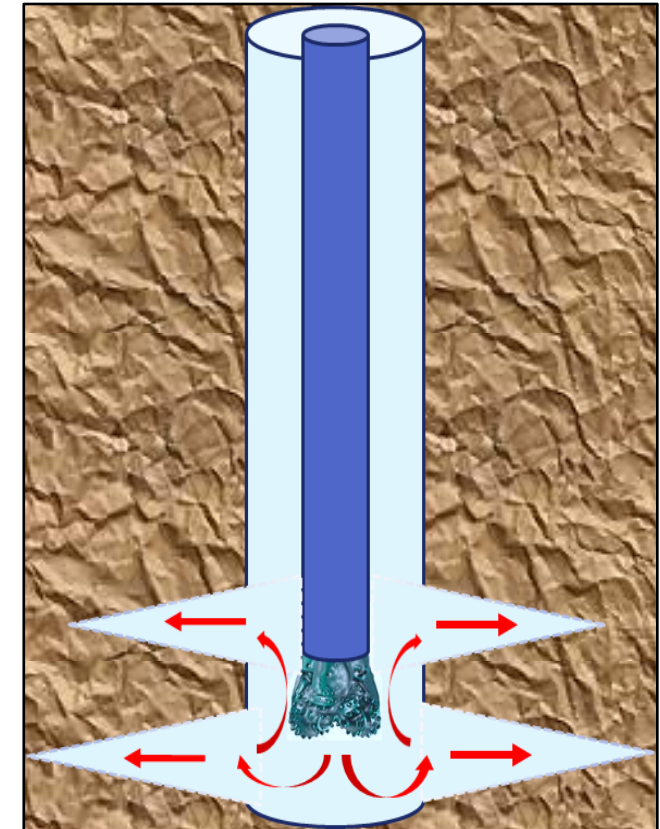
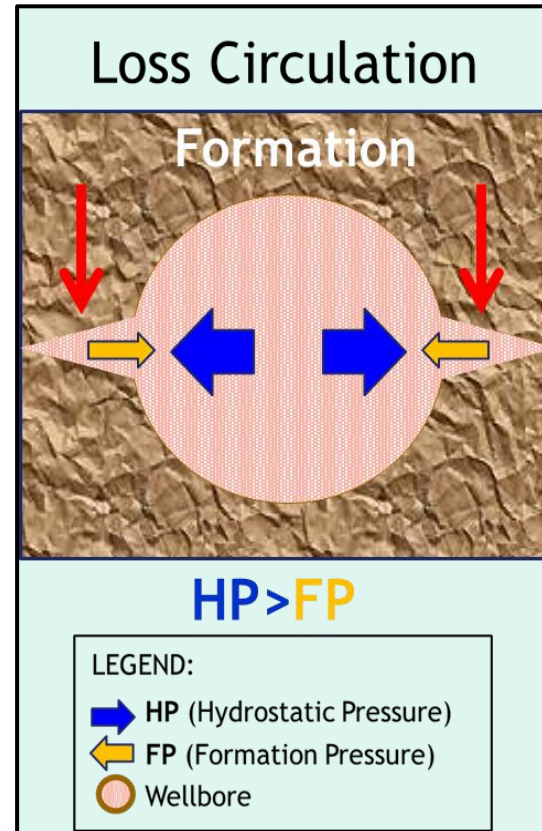
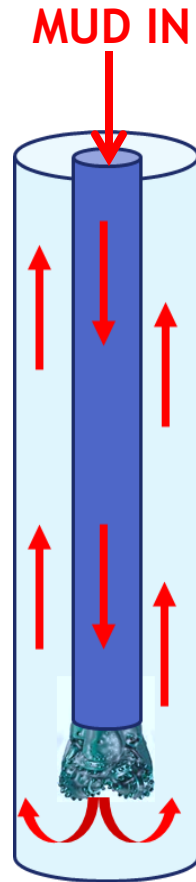
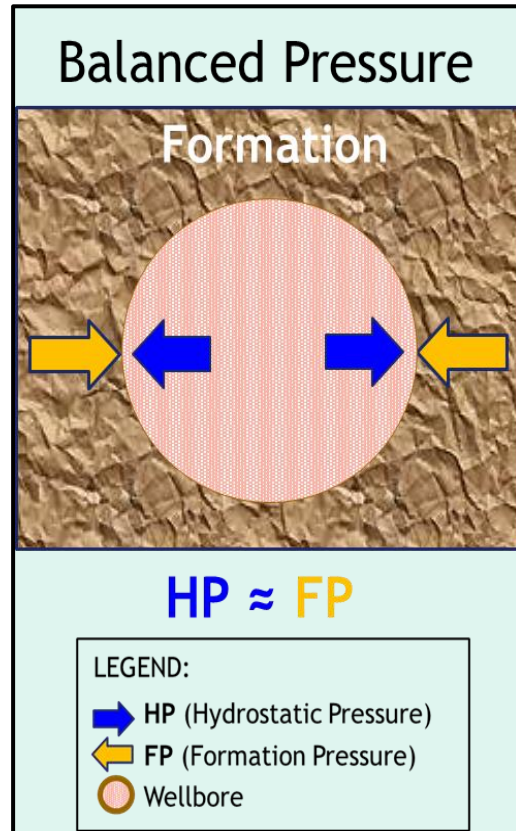
# Stuck Pipe (Pipe Sticking)

Hydrostatic Pressure (HP)  $>$  Formation Pressure (FP)



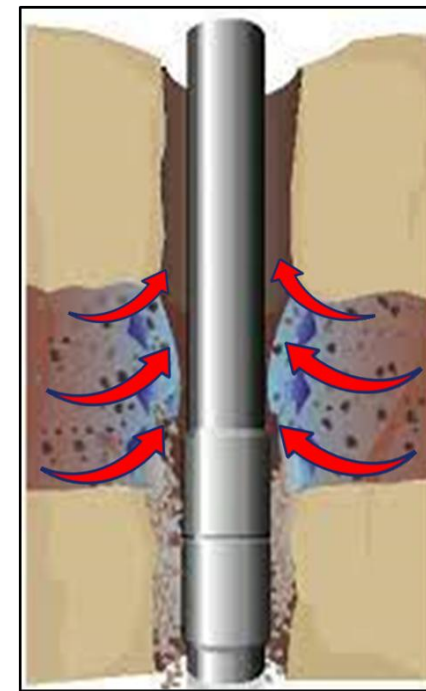
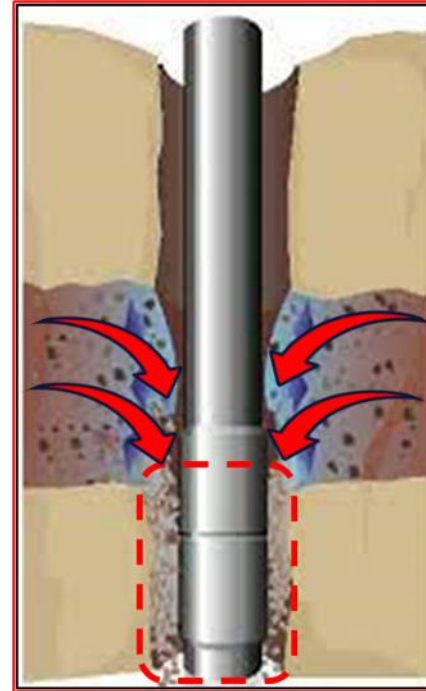
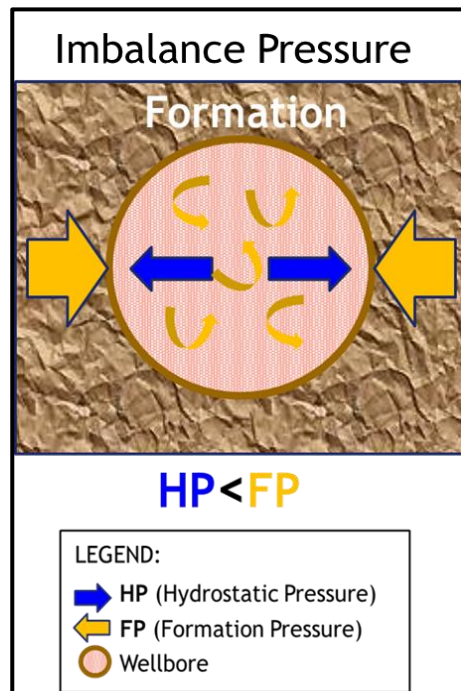
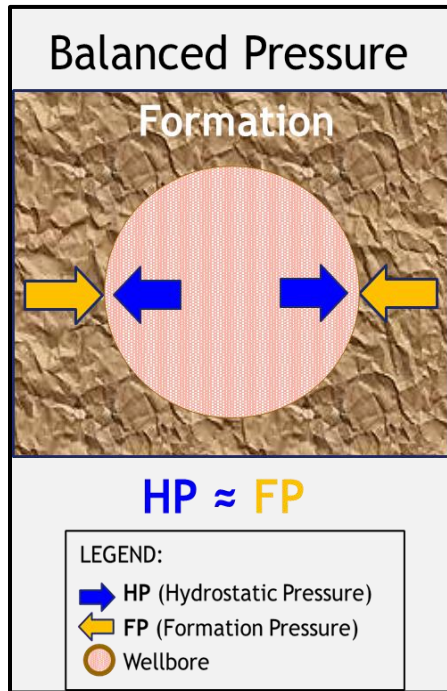
# Loss Circulation

Hydrostatic Pressure (HP)  $>$  Formation Pressure (FP)



# Gas Kick (Blow Out)

Hydrostatic Pressure (HP)  $<$  Formation Pressure (FP)



# Collect Subsurface Geological Information

## 3. Collect Subsurface geological materials: “drilled cuttings”



# Typical Lithology Descriptions

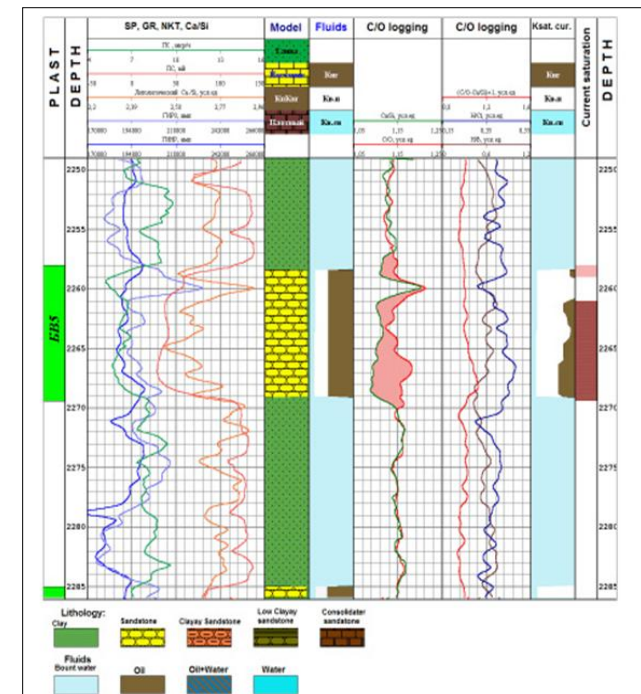
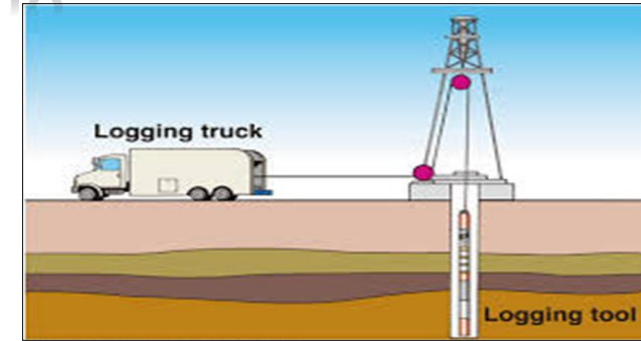
| Lithology | %   | Description                                                                                               |
|-----------|-----|-----------------------------------------------------------------------------------------------------------|
| Claystone | 100 | Medium light grey, greenish grey, <b>moderately hard</b> , strongly calcareous.                           |
| Claystone | 30  | Medium light grey, greenish grey, <b>soft and sticky</b> , non to slightly calcareous                     |
| Sand      | 70  | Yellowish grey, very light grey in part, <b>loose, coarse</b> , angular grain, <b>trace coarse pyrite</b> |
| Claystone | 100 | Medium light grey, greenish grey, <b>soft and sticky, silty in part</b> , non to slightly calcareous      |
| Claystone | 80  | Medium light grey, greenish grey, <b>soft and sticky</b> , non to slightly calcareous                     |
| Sand      | 20  | Very light grey in part, <b>loose, coarse</b> , angular grain, <b>trace coarse pyrite</b>                 |



# Electric Wireline Logging

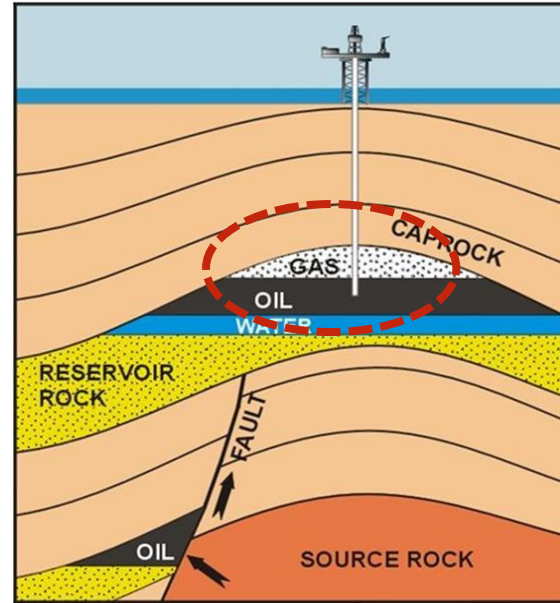
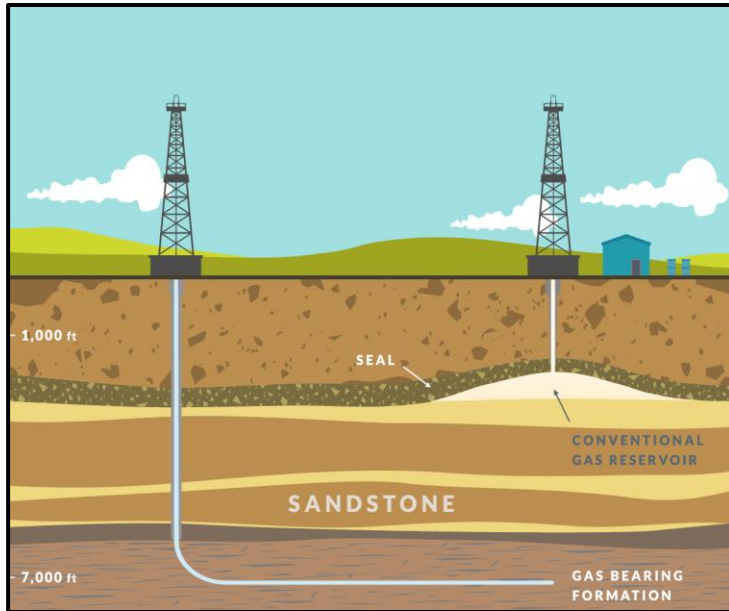
## 4. Interpreting electric logs:

- \* Lithology of formation
- \* Actual Wellbore Size
- \* Temperature
- \* Any Hydrocarbon??

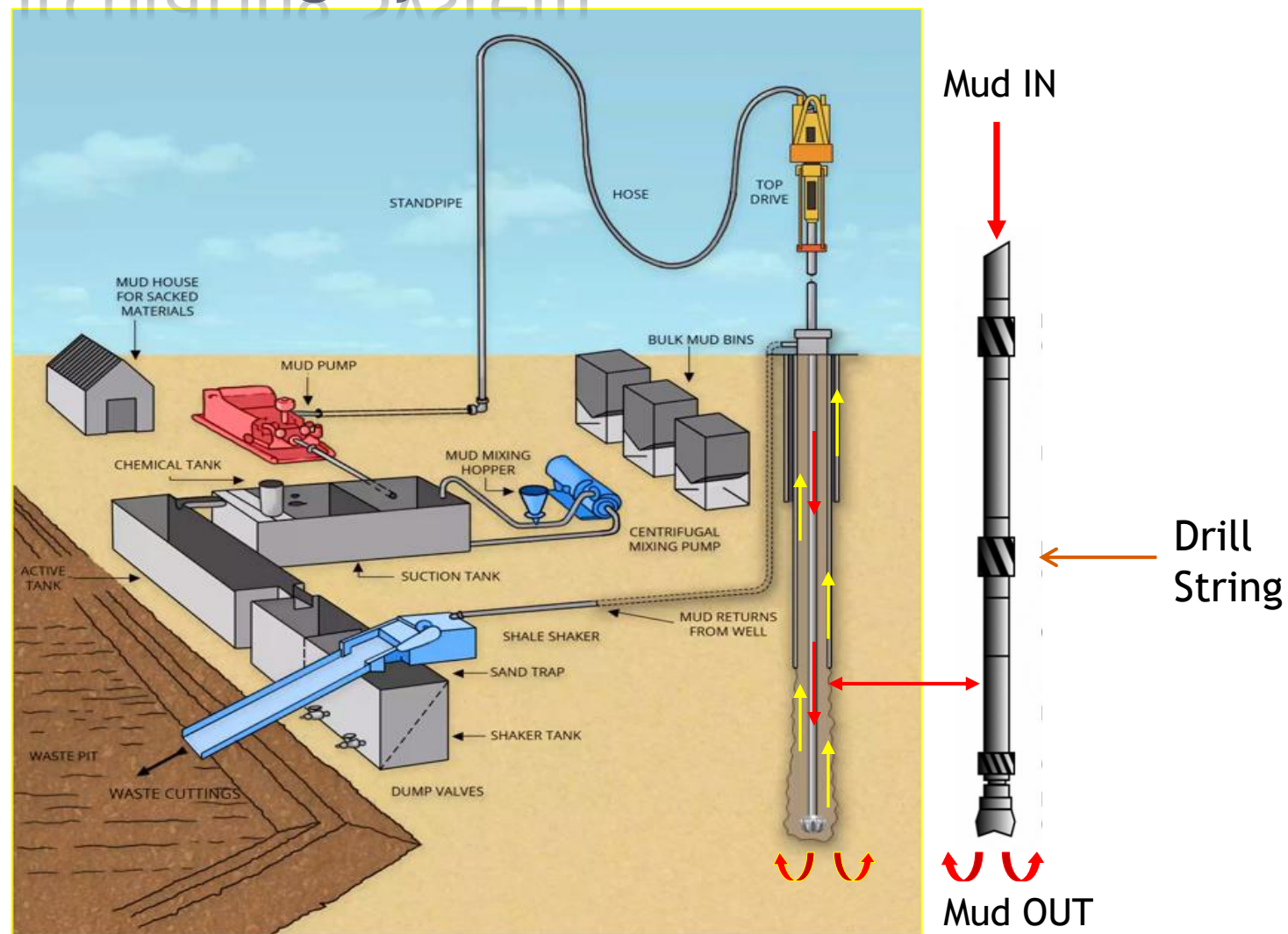
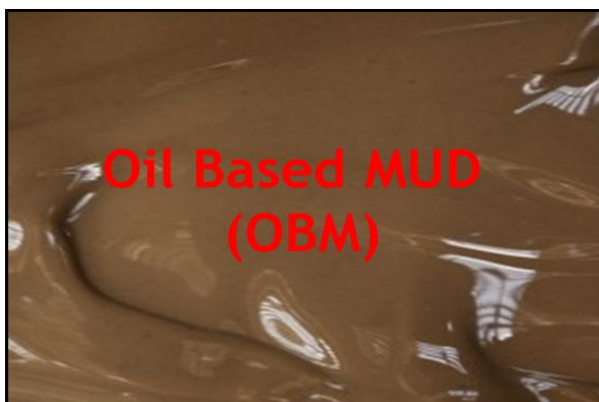


# Minimize Reservoir Damage

## 5. Minimize reservoir damage



# Circulating System



# Drilling Fluid Classification



| Drilling Fluids Classification |               |               |
|--------------------------------|---------------|---------------|
| Pneumatic Fluids               | WBM           | OBM           |
| Air                            | Dispersed     | Diesel Oil    |
| Foam                           | Non-Dispersed | Mineral Oil   |
| Aerated                        | -             | Synthetic Oil |
| -                              | -             | Ester         |



# Pneumatic Fluids

Application: For drilling depleted or low pressure zones.

| Advantages                                                                                                                                                            | Disadvantages                                                                                                                                                            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>* Drill fast</li><li>* Minimize formation damage</li><li>* No clay hydration</li><li>* No precipitation in pore space</li></ul> | <ul style="list-style-type: none"><li>* Need extra rig equipment: air/gas compressor, etc</li><li>* Ineffective in areas of large volumes of formation fluids.</li></ul> |



# Water-Based Mud Systems

| DISPERSED               |                                                                                                           | NON-DISPERSED                                                                                                                     |                                                                                      |
|-------------------------|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Dispersed/<br>Inhibited | Dispersed/<br>Non-Inhibited                                                                               | Non-Dispersed/<br>Inhibited                                                                                                       | Non-Dispersed/<br>Non-Inhibited                                                      |
| * Lime Mud              | <ul style="list-style-type: none"> <li>* Lignosulfonate</li> <li>* Lignite/<br/>Lignosulfonate</li> </ul> | <ul style="list-style-type: none"> <li>* KCl/Polymer</li> <li>* NaCl/Polymer</li> <li>* HPWBM</li> <li>* Formate-based</li> </ul> | <ul style="list-style-type: none"> <li>* Bentonite</li> <li>* Gel/Polymer</li> </ul> |

- \* Dispersed: disperse formation/drilled cuttings
- \* Non-dispersed: inability to disperse formation/drilled cuttings
- \* Inhibited: ability to inhibit swelling of clays/shales
- \* Non-Inhibited: inability to inhibit swelling of clays/shales



# Water-Based Muds

## DISPERSED MUD SYSTEMS



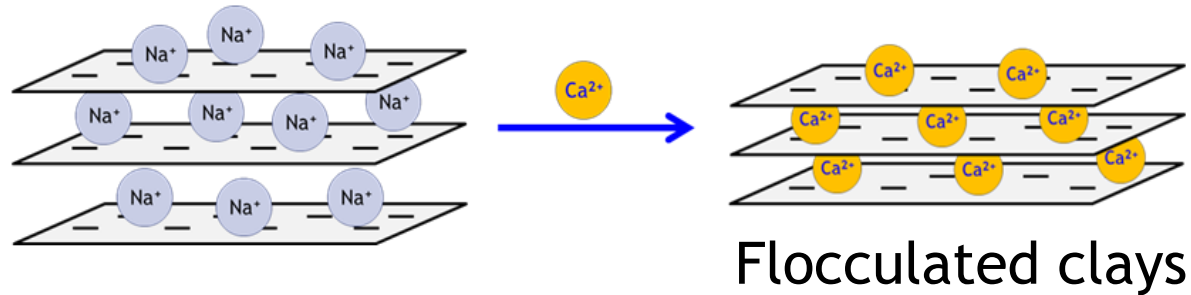
# DISPERSED Mud System

- \* Mud system: Dispersants-based mud
- \* Usage: drilling reactive clays formation
- \* Typical mud systems:
  - \* **Dispersed/Inhibited**: Lime mud system
  - \* **Dispersed/Non-Inhibited**:
    - \* Lignosulfonate mud system
    - \* Lignite/Lignosulfonate mud system



# Dispersed/Inhibited System: Lime Mud

- \* Primary product: Lime:  $\text{Ca}(\text{OH})_2$
- \* The inhibition occurs when  $\text{Ca}^{2+}$  ions replace  $\text{Na}^+$  ions in clay layers ~ resulting in flocculation
- \* Usage: drilling reactive clay formation
- \* Flocculated clays will precipitate, reduce rheological properties.



# Lime Mud System: Typical Products Usage

| Generic         | Function         | Remarks         |
|-----------------|------------------|-----------------|
| Water           | Base fluid       | Freshwater      |
| Bentonite       | Viscosifier      |                 |
| Caustic soda    | Alkalinity       | Optional        |
| Lignosulfonate  | Thinner          |                 |
| Lignite         | Thinner          | For higher BHT  |
| Lime            | Clay inhibitor   | Primary product |
| Starch or PAC   | Fluid loss agent |                 |
| Barite          | Weighting agent  |                 |
| Other additives | -                | As needed       |

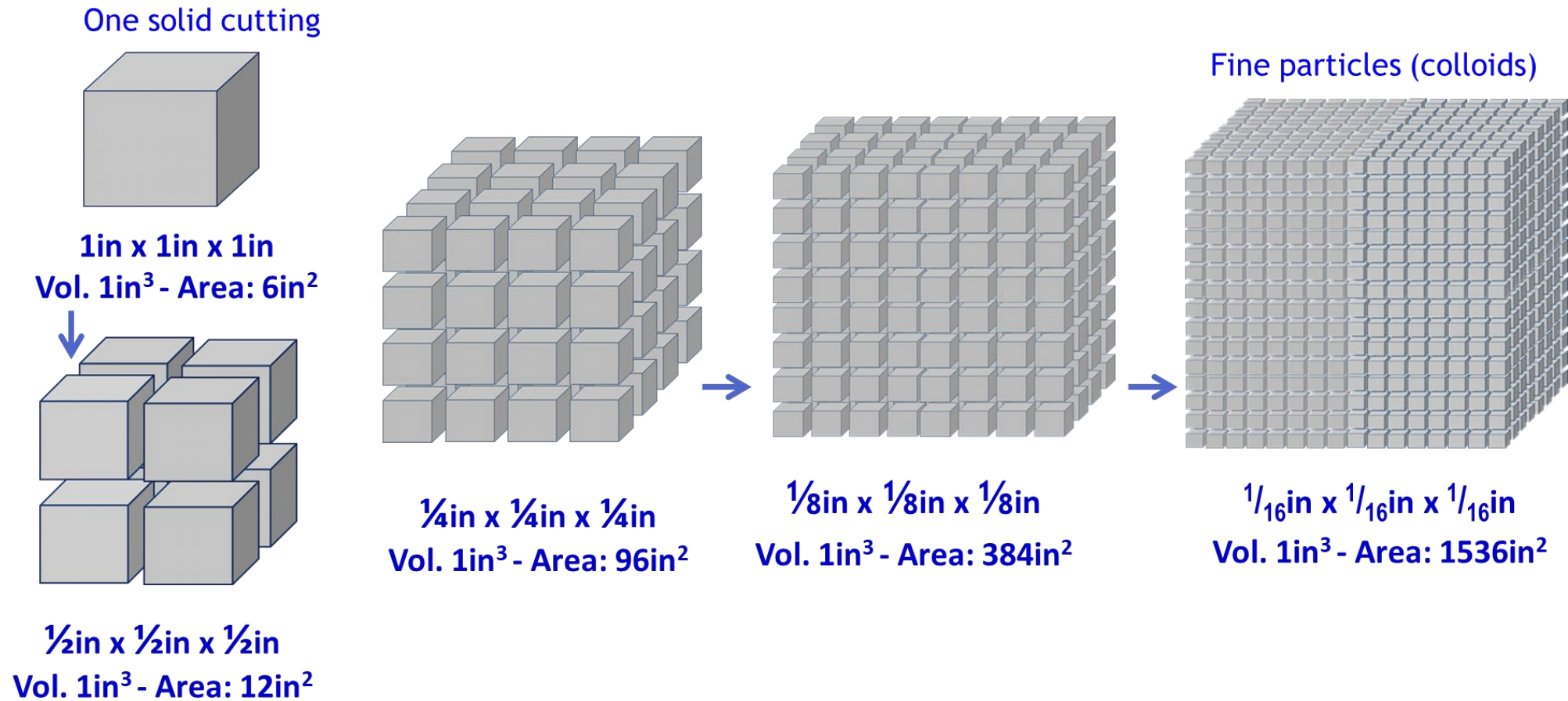


# Dispersed/Non-Inhibited System: Lig./Lignosulfonate

- \* Primary product: Lignosulfonate, Lignite
- \* Usage: drilling reactive clay formation
- \* Disperse reactive clay formation and drilled cuttings, reduce rheological properties; create more surface areas.
- \* Require good solids control equipment
- \* Not recommended for drilling reservoir



# Dispersed System: Performance Mechanism



# Lignite/Lignosulfonate System: Typical Products Usage

| Generic         | Function         | Remarks                    |
|-----------------|------------------|----------------------------|
| Water           | Base fluid       | Normally fresh water       |
| Bentonite       | Viscosifier      | As needed                  |
| Caustic soda    | Alkalinity       | Always maintain pH 9.5-9.7 |
| Lignosulfonate  | Thinner          | pH 4.5-5.0                 |
| Lignite         | Thinner          | For higher BHT             |
| Starch or PAC   | Fluid loss agent |                            |
| Barite          | Weighting agent  |                            |
| Other additives | -                | As needed                  |



# Drilling Fluid Systems

## Water-Based Muds

### NON-DISPERSED MUD SYSTEMS



## NON-DISPERSED/Inhibited Mud System

- \* Mud system: Polymer-based mud
- \* Usage: drilling reactive clays formation
- \* Typical mud systems:
  - \* Freshwater/Polymer, Seawater/Polymer
  - \* KCl-NaCl/Polymer
  - \* KCl-NaCl/Polymer/PHPA/Glycol
  - \* High Performance WBM (HPWBM)
- \* Other systems: Potassium Formate, Potassium Acetate, Potassium Sulfate.



## Non-Dispersed/Inhibited System: Typical Products Usage

| Generic                            | Function                              | Remarks              |
|------------------------------------|---------------------------------------|----------------------|
| Water                              | Base fluid                            | Fresh or salty water |
| Xanthan Gum                        | Viscosifier                           |                      |
| Caustic soda                       | Alkalinity                            | pH 9.0-9.5           |
| Calc Carbonate F, M<br>Marble F, M | F-Weighting agent<br>M-Bridging agent | Marble for HT        |
| Starch or PAC                      | Fluid loss agent                      |                      |
| KCl, NaCl, Polyamine               | Clay inhibitor                        |                      |
| Barite                             | Weighting agent                       | Contingency          |
| Other additives                    | -                                     | PHPA, Glycol, etc.   |



# NON-DISPERSED/Non-Inhibited Mud System

- \* Mud system: Polymer-based mud
- \* Usage: drilling non-reactive clays formation
- \* Typical mud systems:
  - \* Bentonite system: spud mud system
  - \* Bentonite/Polymer system
  - \* Polymer system



# NON-DISPERSED / Non-Inhibited : Typical Products Usage

Mud system: Spud mud

| Generic         | Function        | Remarks              |
|-----------------|-----------------|----------------------|
| Water           | Base fluid      | Fresh or salty water |
| Bentonite       | Viscosifier     | Spud mud             |
| Caustic soda    | Alkalinity      | pH 9.5-9.7           |
| Barite          | Weighting agent | Contingency          |
| Other additives | -               | As needed            |



## NON-DISPERSED / Non-Inhibited : Typical Products Usage

| Generic                            | Function                              | Remarks     |
|------------------------------------|---------------------------------------|-------------|
| Water                              | Base fluid                            | Freshwater  |
| Bentonite                          | Viscosifier                           | Spud mud    |
| Xanthan Gum                        | Viscosifier                           |             |
| Caustic soda                       | Alkalinity                            | pH 9.0-9.5  |
| Calc Carbonate F, M<br>Marble F, M | F-Weighting agent<br>M-Bridging agent | Optional    |
| Starch or PAC                      | Fluid loss agent                      |             |
| Barite                             | Weighting agent                       | Contingency |
| Other additives                    | -                                     |             |

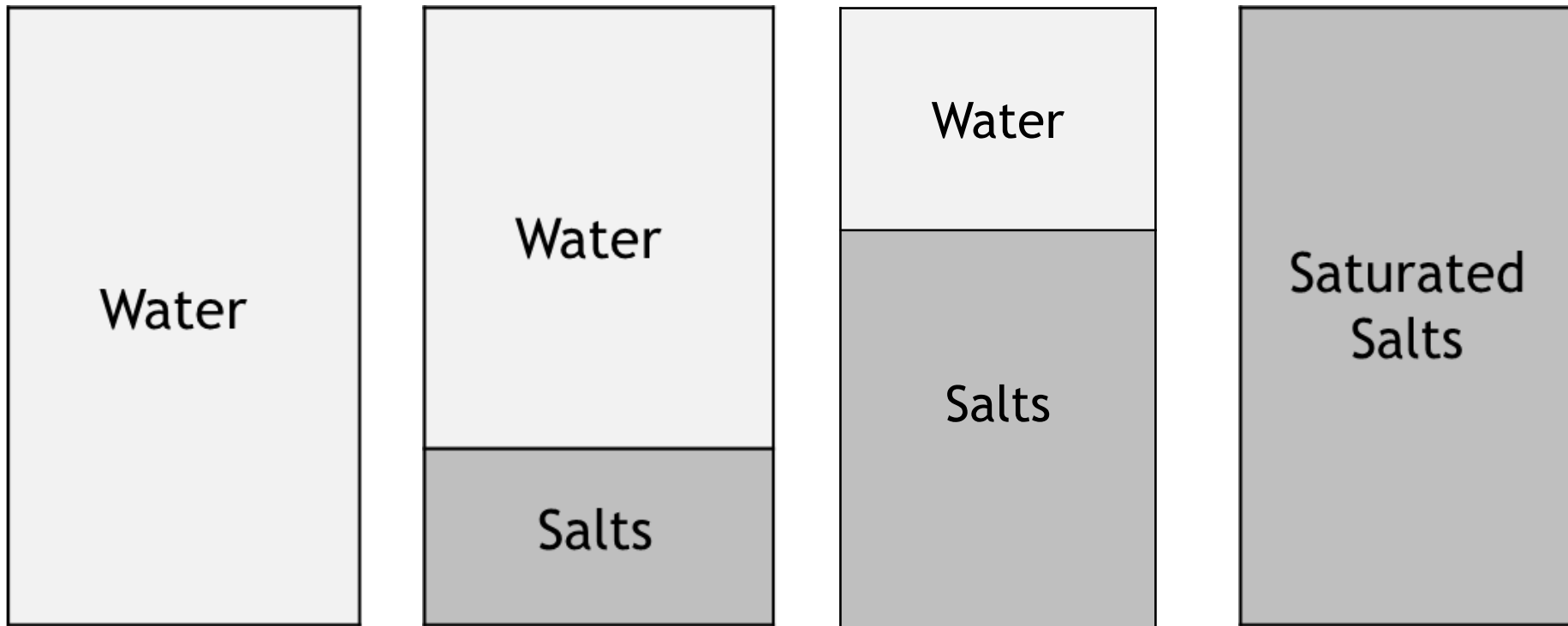


# WBM Systems Application: Reservoir/Non-Reservoir

| Description   | DISPERSED<br>( <i>Lime, Lignite, Lignosulfonate</i> ) |                                                         | NON-DISPERSED<br>( <i>Polymer-based</i> ) |                                         |
|---------------|-------------------------------------------------------|---------------------------------------------------------|-------------------------------------------|-----------------------------------------|
|               | Inhibited<br>( <i>Lime</i> )                          | Non-Inhibited<br>( <i>Lignite,<br/>Lignosulfonate</i> ) | Inhibited<br>( <i>KCl, NaCl</i> )         | Non-Inhibited<br>( <i>No KCl/NaCl</i> ) |
| RESERVOIR     | Optional                                              |                                                         | Optional                                  | Optional                                |
| NON-RESERVOIR | ✓                                                     | ✓                                                       | ✓                                         | ✓ ( <i>spud mud</i> )                   |



# Salts-Based Solutions



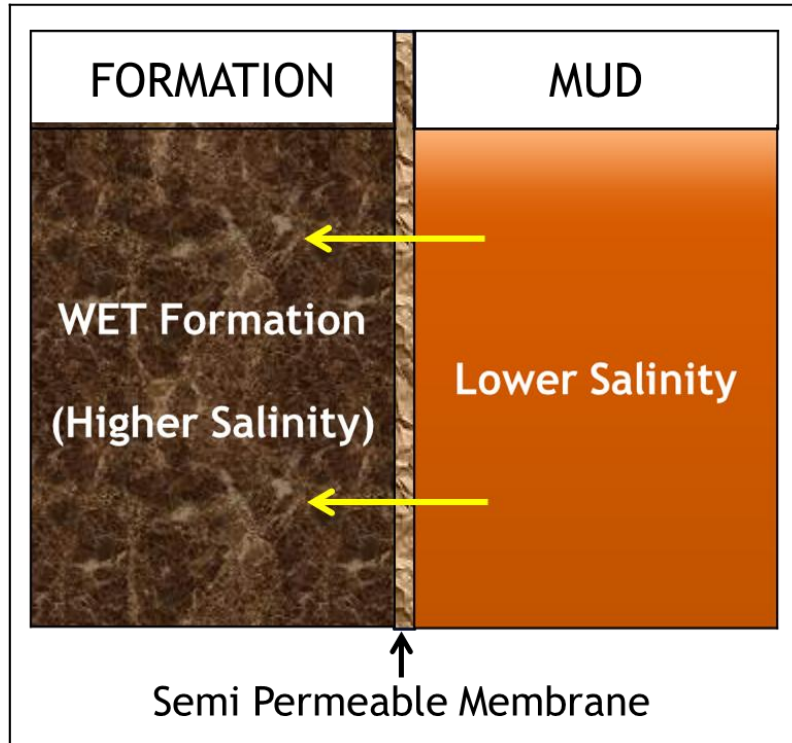
*Note:*

Always require free water to allow maximum polymers reaction.

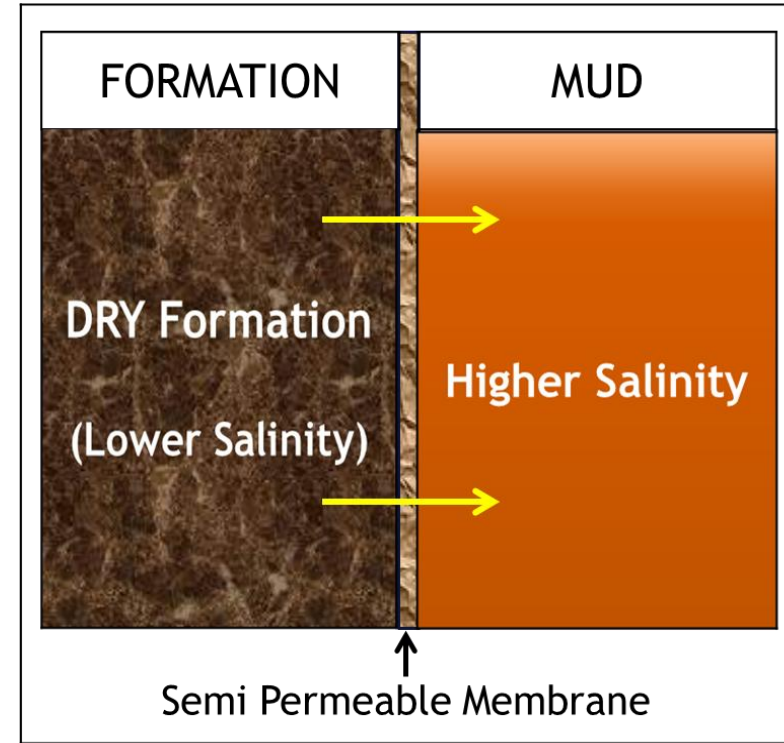


# Inhibition Mechanism

## Hydration (Swelling)



## Dehydration



# High Performance WBM (HPWBM) Highly Inhibitive Aqueous Fluid (HIAF)



# HPWBM: The Benefits

- \* Replacement for Chloride-based system
- \* Stabilize highly reactive clay formations
- \* Designed for drilling reservoir
- \* HTHP stability
- \* Environmentally friendly
- \* Ability to control reactive clays is as good as an OBM system



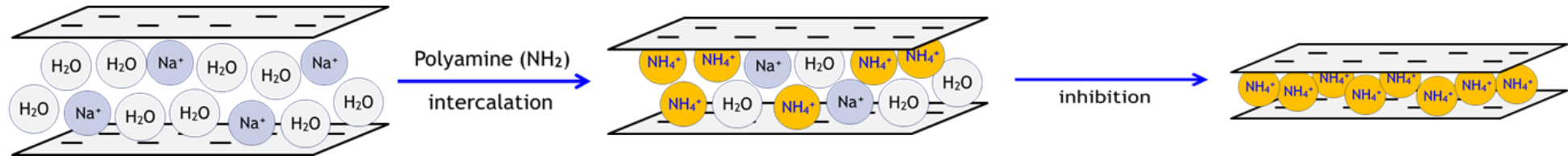
# HPWBM: Typical Products Usage

| Material                            | Function                 | Remarks      |
|-------------------------------------|--------------------------|--------------|
| Starch, PAC                         | Fluid Loss agent         | PAC optional |
| Polyamine                           | Shale inhibitor          | Primary      |
| Accretion suppressant               | Anti accretion           |              |
| Sealing agent                       | Sealing micro-fractures  |              |
| CaCO <sub>3</sub> F/M or Marble F/M | Weighting/bridging agent |              |
| Xanthan Gum                         | Viscosifier              |              |
| KCl, NaCl                           | Clay inhibitor           | KCl optional |
| Caustic soda                        | pH control               | optional     |
| Barite                              | Weighting agent          | n/a          |
| Other additives                     | LCM, lubricant, etc      | As needed    |



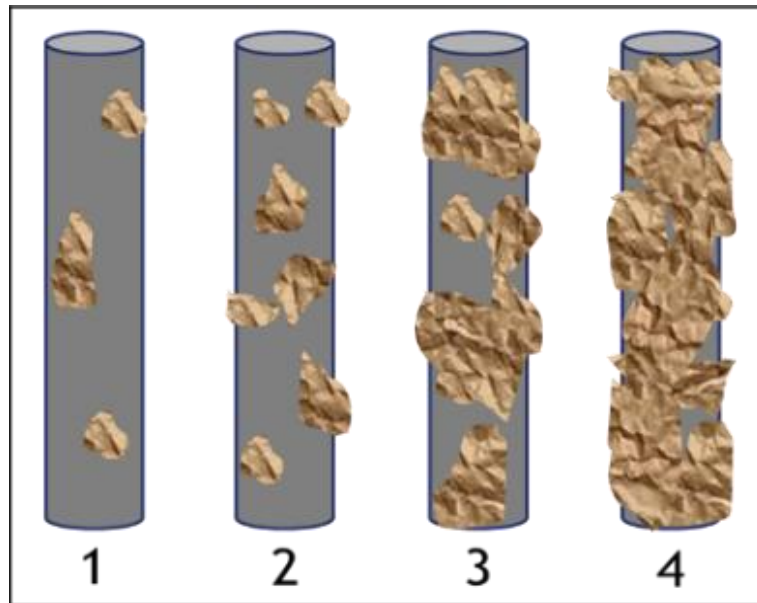
# Polyamine: Performance Mechanism

The adsorption of polyamine onto clay particles makes the clay more *hydrophobic*, prevent water penetration into clay.



# Accretion Suppressant

- \* Inhibit water penetration into the clay and bind platelets together.
- \* Coat BHA, preventing from adhering to metal surface.



1. Minimum amount of sticky clays - good inhibition treatment
2. Slight increased of sticky clays - minimum inhibition treatment
3. Increased sticky clays - insufficient treatment
4. No inhibition treatment



# Formate Based Mud Systems



# Formate Mud Systems: The Benefits

- \* Replacement for Chloride-based fluid systems
- \* Stabilize highly reactive clays
- \* Formulate Low Solids Mud system
- \* Extend thermal stability of polymers
- \* Less corrosion
- \* HTHP stability
- \* Environmentally friendly



# Formate Products

| Generic Names                | Max. Density,<br>ppg (SG) | Max Temp.<br>°C (°F) | Remarks           |
|------------------------------|---------------------------|----------------------|-------------------|
| Sodium Formate<br>(NaCOOH)   | 11.10 (1.33)              | 165° (329°)          | Powder            |
| Potassium Formate<br>(KCOOH) | 13.20 (1.58)              | 165° (329°)          | Powder/<br>Liquid |
| Cesium Formate<br>(CsCOOH)   | 19.20 (2.30)              | 205° (400°)          | Liquid            |



# Formate-based System: Typical Products Usage

| Material                               | Function                                                                                                            | Remarks       |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------|---------------|
| Water                                  | Base fluid                                                                                                          |               |
| Potassium Formate                      | <ul style="list-style-type: none"> <li>• Base fluid</li> <li>• Weighting agent</li> <li>• Clay inhibitor</li> </ul> | Primary       |
| CaCO <sub>3</sub> F/M or Marble F/M    | Weighting/bridging agent                                                                                            | Marble for HT |
| Xanthan Gum                            | Viscosifier                                                                                                         |               |
| Starch, PAC                            | Fluid Loss agent                                                                                                    | PAC optional  |
| Manganese Tetroxide,<br>Barite, others | Weighting agent                                                                                                     |               |
| Other additives                        | LCM, lubricant, etc                                                                                                 | As needed     |



# Deep Water Mud Systems



# Deep Water Mud System

Primary concerns in Drilling Fluid design are:  
Gas Hydrates and Low flowline temperature.



# Gas Hydrates

- \* A crystalline solids, ice-like, mixture of gas and water.
- \* Occur naturally at temperature of  $<50^{\circ}$  F; water temperature decreases with depth.
- \* Require high salinity to suppress hydrates; deeper well needs higher salinity.
- \* OBM provides excellent hydrate suppression due to low water content that has high salinity.



# Water-Based Mud: Degree of Inhibition

Inhibition  
less  
 ↓  
more

| Mud System                | Reservoir | Non-Reservoir |
|---------------------------|-----------|---------------|
| Polymer                   | √         | √             |
| Polymer/PHPA              | √         | √             |
| KCL/Polymer               | optional  | √             |
| KCL/Polymer/PHPA          | optional  | √             |
| Polymer/PHPA/Glycol       | √         | √             |
| KCL/Polymer/PHPA/Glycol   | optional  | √             |
| HPWBM, Formate-based, etc | √         | √             |



# Oil Based Mud (OBM) Non-Aqueous Fluid (NAF)



# QBM: Benefits

- \* When WBM system is ineffective
- \* Effective in drilling highly reactive clays/shale, loose sands
- \* Resist to high temperature (HT)
- \* Contamination resistance: cement, anhydrate,  $H_2S$ ,  $CO_2$
- \* Corrosion resistance
- \* Excellent lubricity
- \* Minimize potential bit balling, tight hole and/or stuck pipe
- \* Near-gauge wellbore



# OBM: Benefits

- \* Production protection
- \* Perform better to drill highly deviated, horizontal hole section
- \* Zero discharge (e.g. ester based)
- \* Re-usable



# OBM: Limitations

- \* Expensive if only drill 1-2 wells.
- \* Environmental concern if using low quality-based oil.
- \* Expensive if use on wells of loss of circulation.
- \* Health concerns.
- \* Requiring special rig equipment.
- \* Considering waste disposal.



# OBM: Based Oils

- \* Petroleum based:
  - \* Crude oil
  - \* Diesel oil
  - \* Mineral oil
- \* Non-petroleum based (organic fluids):
  - \* Synthetic: Ester



# OBM Classification

1. Invert Emulsions

2. All-oil System

| Description      | Water | Salts | Emulsifier | Filtrate |
|------------------|-------|-------|------------|----------|
| Invert Emulsions | yes   | yes   | yes        | all oil  |
| All-Oil system   | no    | no    | a little   | n/a      |



# OBM: Key Success

- \* Primary: mixing order.
  - \* Stable mud properties
  - \* Less treatment, cost-effective
- \* Good shearing facilities
- \* Avoid having water contamination
- \* Good performance of Solids Control Equipment
- \* Mud Engineers' capability

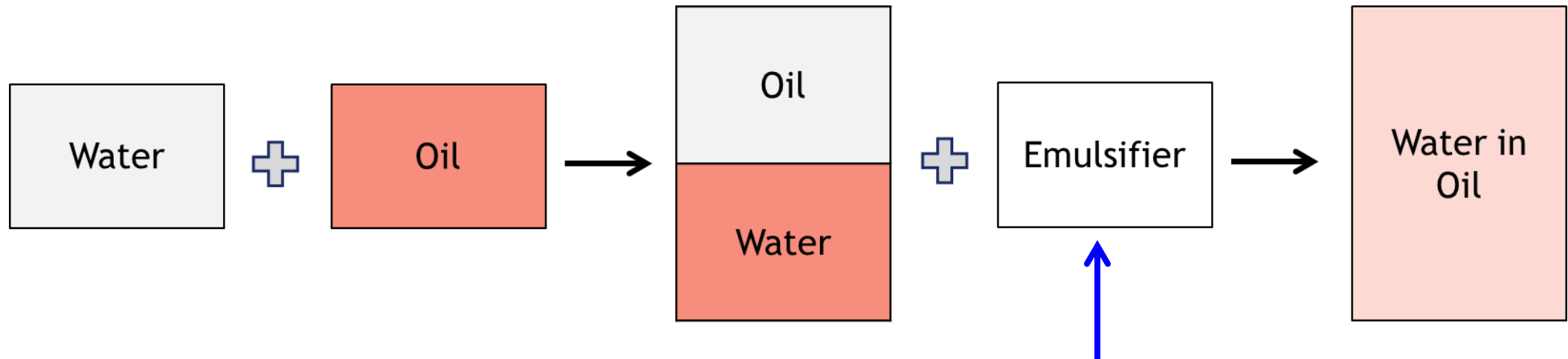


# OBM System: Typical Products Usage

| Material                        | Function           | Remarks                |
|---------------------------------|--------------------|------------------------|
| Oil                             | Base fluid         | Synthetic (preferable) |
| Primary emulsifier              | Emulsifier         |                        |
| Secondary emulsifier            | Emulsifier         | Optional               |
| CaCO <sub>3</sub> F or Marble F | Weighting agent    |                        |
| Organophilic Clay               | Viscosifier        |                        |
| Sulfonated asphalt              | Fluid Loss agent   |                        |
| Lime                            | Alkalinity control | Activate emulsifier    |
| Rheological modifier            | Control rheology   |                        |
| Wetting agent                   | Wetting agent      |                        |
| Other additives                 | LCM, etc           | As needed              |



# Water and Oil Mixture



## Other Fluid Systems

- \* Completion/Work-over Fluids (CWOFF)
- \* Coring Fluids
- \* Milling Fluids
- \* Packer Fluids



# COMPLETION/WORK-OVER FLUIDS (CWOFF)



# CWOF Types

Two basic types CWOF systems:

1. Clear-FLUID systems
2. Solids-enhanced systems:
  - \* WBM
  - \* OBM

Preferred system: Clear-FLUID system which is BRINE, a solids free solution.



# Single SALTS Brine

| Description       | KCl                                                                                                                   | NaCl            | CaCl <sub>2</sub>          | NaBr                                     | CaBr <sub>2</sub> | KCOOH                              | NaCOOH | CsCOOH    |
|-------------------|-----------------------------------------------------------------------------------------------------------------------|-----------------|----------------------------|------------------------------------------|-------------------|------------------------------------|--------|-----------|
| Application       | Water sensitive formation                                                                                             |                 |                            | If large amt of CO <sub>2</sub> presence |                   | If chloride solution is prohibited |        |           |
| Purity            | Min.<br>96%                                                                                                           | ≥ 98%<br>74-78% | 94-97%<br>89-92%<br>77-80% | 100%                                     | 95%               | 100%                               |        |           |
| Density, ppg max. | 9.6                                                                                                                   | 10.0            | 11.6                       | 12.4                                     | 15.1              | 13.3                               | 11.1   | 19.2      |
| Corrosion rate    | Low                                                                                                                   |                 | Med                        | Low                                      | Med               | None                               |        |           |
| Remarks           | <ul style="list-style-type: none"> <li>* Inexpensive</li> <li>* High quality: Pure Vacuum Dried (PVD) NaCl</li> </ul> |                 | Generate dry heat          | Expensive, prone to precipitation        |                   | Expensive                          |        | Most exp. |



## Two SALTS Brine

| Description         | CaCl <sub>2</sub> /CaBr <sub>2</sub>                                          | CaBr <sub>2</sub> /ZnBr <sub>2</sub>     |
|---------------------|-------------------------------------------------------------------------------|------------------------------------------|
| Application         | If require higher density                                                     |                                          |
| Purity              | CaCl <sub>2</sub> (94-97%);<br>CaBr <sub>2</sub> /ZnBr <sub>2</sub> (95-100%) |                                          |
| Density, ppg (max.) | 14.1 - 15.1                                                                   | Up to 19.2 ppg.<br>In practice: 18.0 ppg |
| Corrosion rate      | Moderate                                                                      | High                                     |
| Remarks             | Expensive, precipitation may occur<br>ZnBr <sub>2</sub> : Corrosive and toxic |                                          |



# Three SALTS Brine

| Description    | $\text{CaCl}_2/\text{CaBr}_2/\text{ZnBr}_2$                                                                                                  |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Application    | If require higher density                                                                                                                    |
| Purity         | $\text{CaCl}_2$ (94-97%);<br>$\text{CaBr}_2/\text{ZnBr}_2$ (95-100%)                                                                         |
| Density, ppg   | Up to 19.2 ppg                                                                                                                               |
| Corrosion rate | High @ $>250^\circ\text{F}$                                                                                                                  |
| Remarks        | Equipment may damage in temp $>300^\circ\text{F}$ ( $149^\circ\text{C}$ ); Precipitation may occur,<br>$\text{ZnBr}_2$ : Corrosive and toxic |



# Filtration Operations

The purpose:

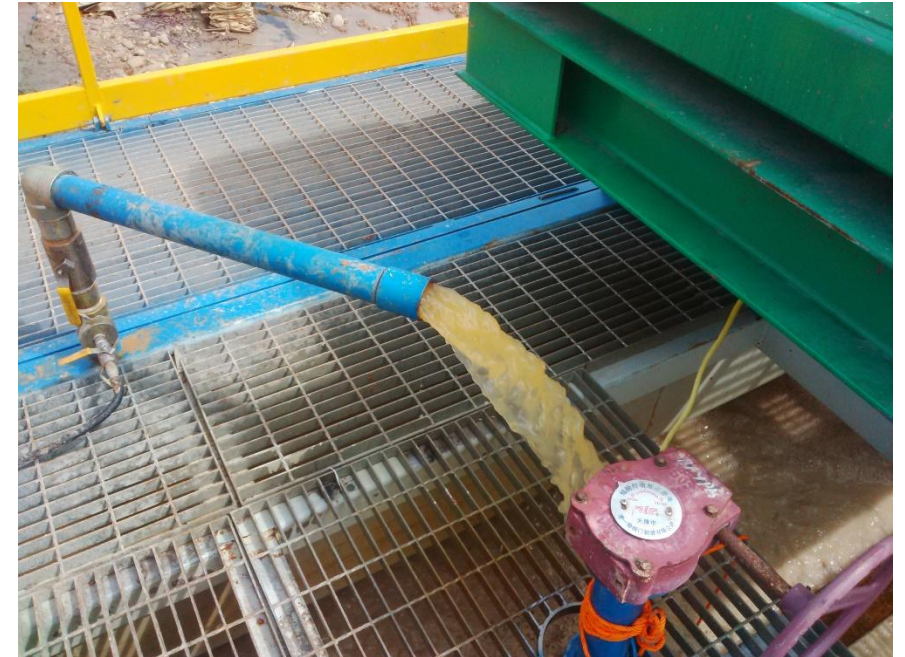
filtering the fluid to remove fine particles/colloids  
to obtain clean and solids-free liquid



# Filtration Operations

Source of particulates:

- \* Dirty water
- \* Bacteria in stagnant water
- \* Products impurities
- \* Sediments
- \* Rust
- \* Remains from mud tanks



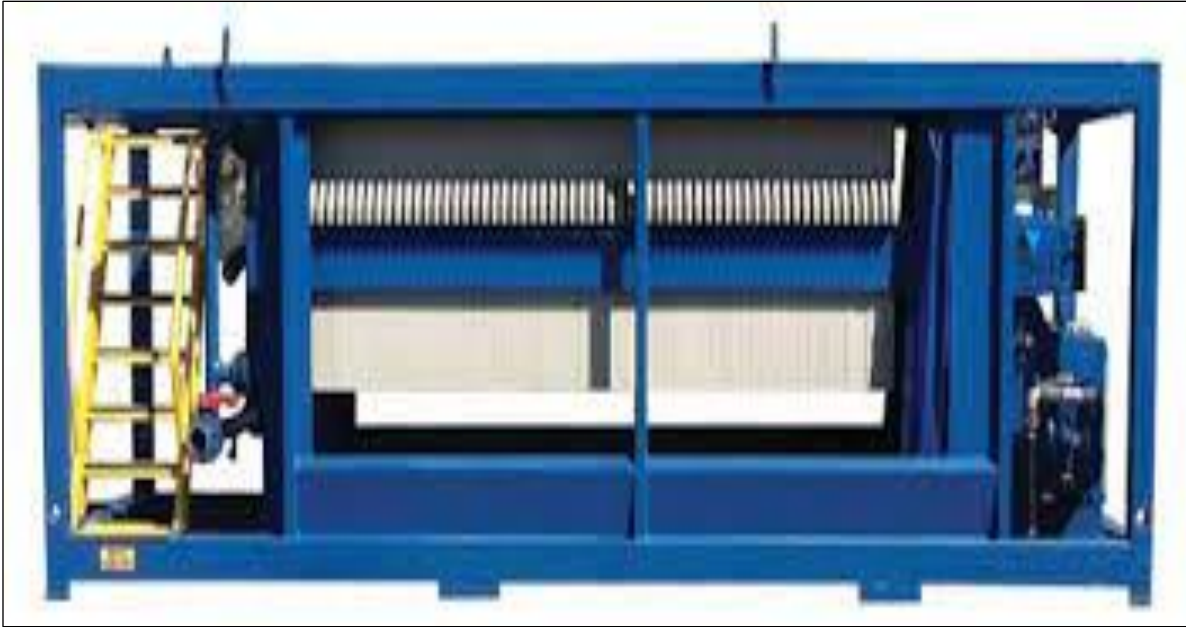
| Filtration Units |                             |                         |
|------------------|-----------------------------|-------------------------|
| Types            | Filtering Media             | Fluid Cleanliness (NTU) |
| DE Unit          | Diatomaceous earth (DE)     | 0 - 2                   |
| Dual Pod Unit    | Cartridges, $\mu$ : 2 to 50 | 10 - 30                 |

*Remarks:*

- \* NTU (Nephelometric Turbidity Unit)
- \* The unit can be used individually or in combination

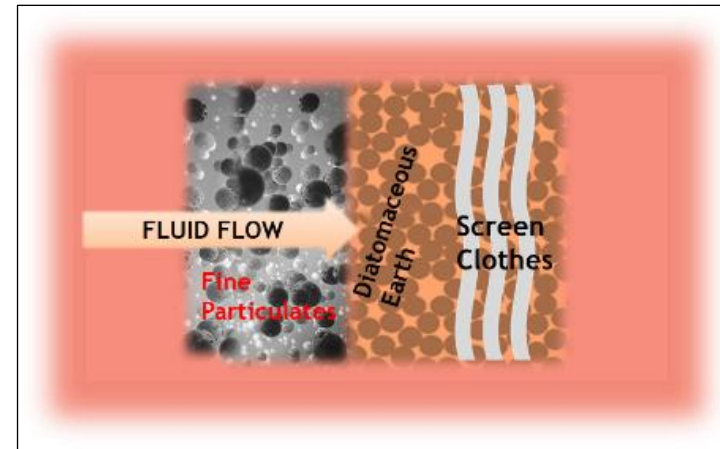


# Filtration Unit: DE Type



- NTU: 0-2
- Filter media: DE powder

Screening process



# Filtration Unit: Dual Pod Unit



- NTU: 10 - 30
- Filter media: filter cartridges of 2, 5, 10, 25, 50  $\mu$

Filter cartridges:

- \* Pleated glass
- \* Wound polypropylene



# Dual Pod Unit

25 or 10  $\mu$

5 or 2  $\mu$



Filter  
cartridges



Before  
Filtering



After  
Filtering



# Fluid Cleanliness

How *clean* does the fluid need to be?

What particle sizes need to be removed?

It is generally accepted that fluids containing particles with diameter of  $<2\mu$  and concentration of  $<100$  mg/L are relatively safe to use as CWOFF.



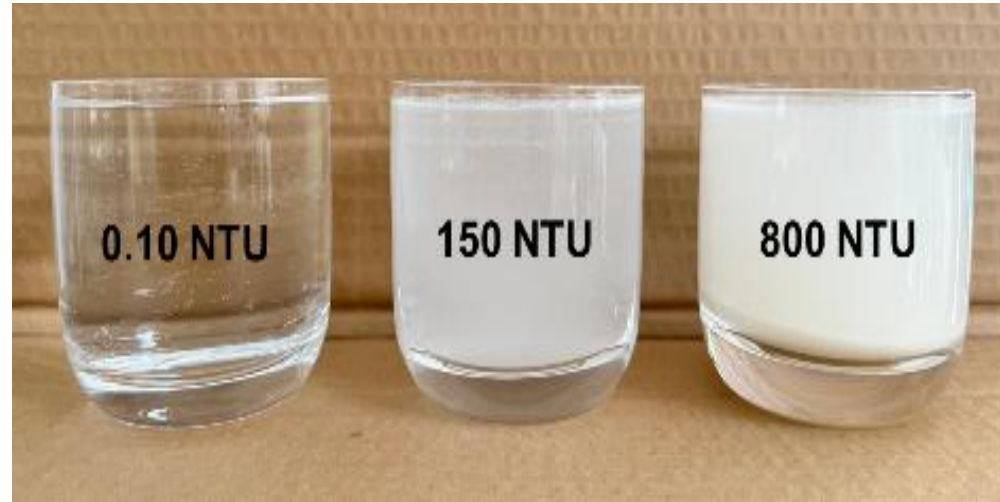
# Fluid Cleanliness - TURBIDITY



- \* Fluid cleanliness is indicated by the turbidity (NTU) of that fluid.
- \* The instrument: Nephelometer, Turbidity Meter or Turbidimeter; measuring concentration of suspended particulates in a liquid.
- \* Unit: NTU (Nephelometric Turbidity Unit)



# Fluid Appearance



- \* Turbidity levels rise or fall as the volume of suspended solids increases or decreases.
- \* Good quality water:  $<1$  NTU
- \* Suspended Solids (SS): 1 mg/l (ppm) equal to 3 NTU.



# Coring Fluid

- \* Designed for Coring
- \* General Fluid specifications:
  - \* DO NOT use fibers material
  - \* Use non-petroleum based lubricant
  - \* Lowest solids content
  - \* Use tracer product



# Milling Fluid

- \* Designed for milling: tubing, drill pipe, casing, etc.
- \* The fluid:
  - \* Use normal mud volume for milling and frequent sweep with high viscosity pills.
  - \* Use mud already formulated with high rheological properties: Yield Point (YP), 6 RPM



# Packer Fluid

- \* A fluid left in hole after completing a well.
- \* The fluid is left static for an extended period of time.
- \* The purpose: to provide hydrostatic pressure to balance pressures between tubing and casing strings.
- \* The fluid: Brine (preferred), WBM, OBM
- \* Packer Fluid should be: Solids free, anti corrosion, non-damaging.



# Subject #3

## Drilling Fluid Properties and Field Testing



# Drilling Fluid Substances

## 1. PHYSICAL substances:

- \* Mud Weight
- \* Viscosity
- \* Rheological properties
- \* ECD
- \* LGS, HGS
- \* etc

## 2. CHEMICAL substances:

- \* pH, Alkalinity
- \* Chloride ( $\text{Cl}^-$ )
- \* Potassium ( $\text{K}^+$ )
- \* Hardness:  $\text{Ca}^{2+}$ ,  $\text{Mg}^{2+}$
- \* etc



Standard Testing Procedure:  
**API Specification 13B-1**  
Recommended Practice for Field  
Testing Water-based Drilling Fluids



# Mud Weight (MW)

- \* Measuring density of Drilling & C/WO Fluids
- \* Equipment: API Mud Balance, Pressurized Mud Balance
- \* Unit: ppg, SG (gr/cc), lb/cuft.



API Mud Balance



Pressurized  
Mud Balance



# Funnel Viscosity

- \* Measuring fluid viscosity
- \* Equipment: Marsh Funnel and Cup
- \* Unit: Sec/qt (Second/quartz); 1 qt = 946 mls.



# Plastic Viscosity

- \* Indicating volume, type and size of solids
- \* Unit: cP (Centipoise), Pa.sec
- \* Value: 600-rpm minus 300-rpm



# Plastic Viscosity

- \* Indicating volume, type and size of solids
- \* Value: 600-rpm minus 300-rpm; unit cP (Centipoise)



**FANN V-G Meter**  
**Model: 35A**



**FANN Viscometer**  
**Model: iX77**



**OFITE Variable**  
**Speed Viscometer**



# Yield Point (Yield Value)

- \* Indicating hole cleaning capability.
- \* Equipment: Rheometer, VG-Meter, Fann ix77
- \* Unit: lb/100 ft<sup>2</sup>, Pa.
- \* Value: Subtract PV value from 300-rpm reading.



# Gel Strength

- \* Attractive forces under static condition.
- \* Develops by time.
- \* Values vary by hole size, MW and well profile
- \* Equipment: Rheometer, VG-Meter, Fann ix77
- \* Unit: lb/100 ft<sup>2</sup>, Pa.
- \* Values recorded in 3-rpm reading at: 10s/10m/30m time interval



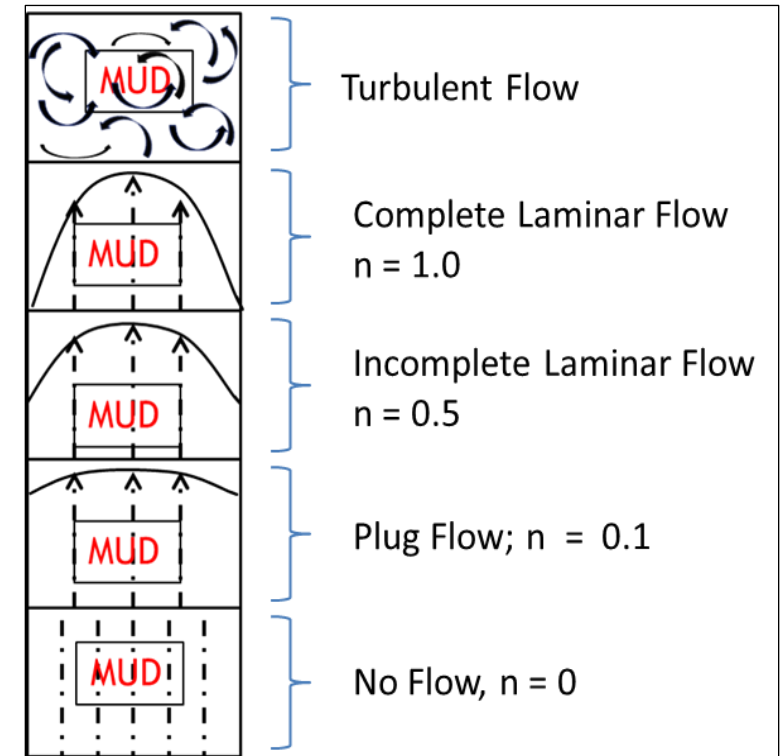
## 6 RPM Reading

- \* Indicating hole cleaning and carrying capacity in low pump rate.
- \* Equipment: Rheometer, VG-Meter, Fann ix77
- \* Unit: lb/100 ft<sup>2</sup>.
- \* Value is generated from 6-rpm reading.



# $n$ - Value

- \*  $n$ : laminar flow index
- \* Indicating mud flowing mode.
- \* Most Polymer mud system has  $n$  value of 0.4-0.6
- \* Result: Calculations, data from VG-Meter, Fann ix77 readings.
- \* Unit: dimensionless



# Low Gravity Solids

- \* An indicator of quality of mud
- \* LGS: drilled solids, products' impurities
- \* Average density: 2.6 gr/cc
- \* Normal value: 50-55 lb/bbl or 5-6% by volume.
- \* Unit: lb/bbl or % by vol.



# High Gravity Solids (HGS)

- \* HGS refers to products with higher specific gravity:  
Barite (sg 4.2), Manganese Tetroxide (sg 4.8), Hematite (sg 5.0).
- \* Inert component
- \* Unit: lb/bbl, % by vol.



# API Fluid Loss

- \* Measuring the rate of fluid loss into formation
- \* Equipment: API Filter Press
- \* Unit: mls/30 min.
- \* Operate in 100 psi, at ambient temperature.



# HTHP Fluid Loss

- \* Measuring Fluid Loss at HT
- \* Observing products resistant to HT
- \* Unit: mls/30 min.
- \* Operating pressures:
  - \* 600 psi on top regulator
  - \* 100 psi on bottom regulator.



HTHP Filter Press



# HTHP Fluid Loss



- \* Equipment: FANN Model 90
- \* Unit: mls/30 min.
- \* Operate in *actual* bottom hole temperature and pressure.



# Filter (MUD) Cake

- \* The residue deposited on a permeable medium that is forced under a pressure
- \* Desirable quality: impermeable, thin and tough.
- \* Desirable thickness:  $\frac{2}{32}$  of an inch.



# pH (Alkalinity)

- \* Measuring pH (alkalinity) of a fluid
- \* pH of most WBM's: 9 - 10
- \* Equipment: pH Meter, pH Paper
- \* Unit: dimensionless



# Hardness

- \* Other term: Hard Water
- \* The test is performed on make-up water before mixing new mud.
- \* Primarily combination of Calcium and Magnesium ions.
- \* Unit: mg/l; reported as mg/L calcium ion



# Chloride Content

- \* Measure the chloride content in make-up water/mud
- \* Sources:
  - \* Potassium Chloride (KCl)
  - \* Sodium Chloride (NaCl)
  - \* Calcium Chloride (CaCl<sub>2</sub>)
  - \* Seawater
  - \* Formation water
- \* Unit: mg/l, ppm



# Potassium Content

- \* Measuring potassium ( $K^+$ ) content in water/mud
- \* Sources:
  - \* Potassium Chloride (KCl)
  - \* Potassium Hydroxide (KOH)
  - \* Potassium Formate (KCOOH), etc.
- \* Unit: mg/l, ppm



Hand-crank  
Centrifuge

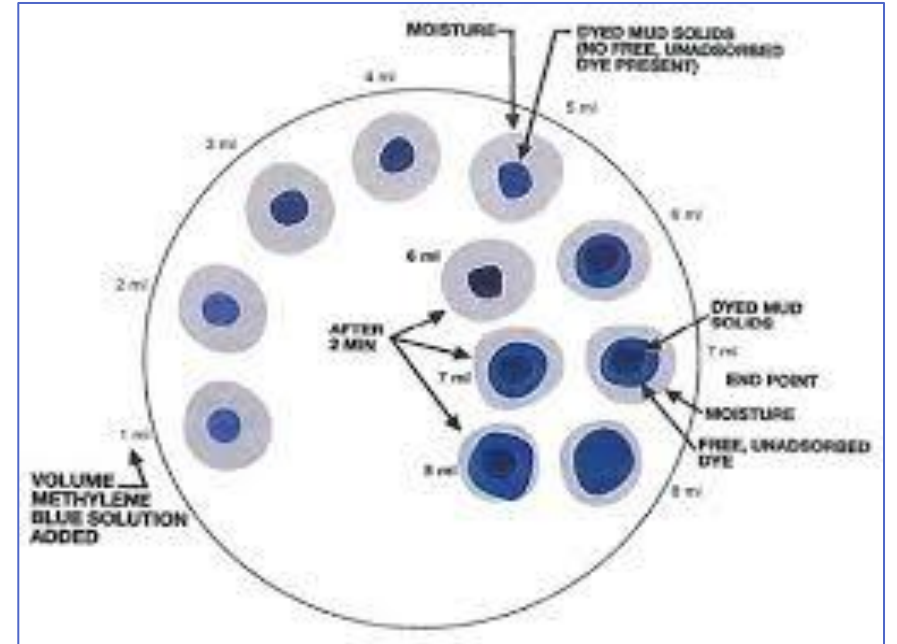


Electric  
Centrifuge



# Methylene Blue Capacity

- \* Indicating amounts of clays in mud system
- \* The value is determined by the Methylene Blue Test (MBT).
- \* Unit: ppb



# Solids, Water, Oil Content



- \* Measuring solids, water and oil content.
- \* Test results are used:
  - \* As guidelines to make treatment
  - \* To justify solids control performance
  - \* To calculate solids content
  - \* In OBM: determine % of oil/water ratio
- \* Equipment: Retort 10 cc, 20 cc, 50 cc.
- \* Unit: % by vol.



**Standard Testing Procedure:**  
**API Specification 13B-2**  
Recommended Practice for Field  
Testing Oil-based Drilling Fluids



# Electrical Stability (ES)

- \* Measuring the emulsion and oil-wetting capability.
- \* Observing oil/water emulsification
- \* Equipment: Electrical Stability (ES) Tester
- \* Unit: Volts



# Salinity

- \* Measuring chloride ( $\text{Cl}^-$ ) content in mud
- \* Needed for inhibition
- \* Hole stabilization.
- \* Unit: mg/l or ppm.



# Excess Lime, Alkalinity

- \* Measuring Lime content in an OBM system
- \* Enhancing performance of emulsifier
- \* Unit: ppb
- \* Excess Lime of 3-6 ppb is desirable.



# Solids Management



## Two type of SOLIDS:

- \* Commercial solids: mud products
- \* Drilled solids from formation



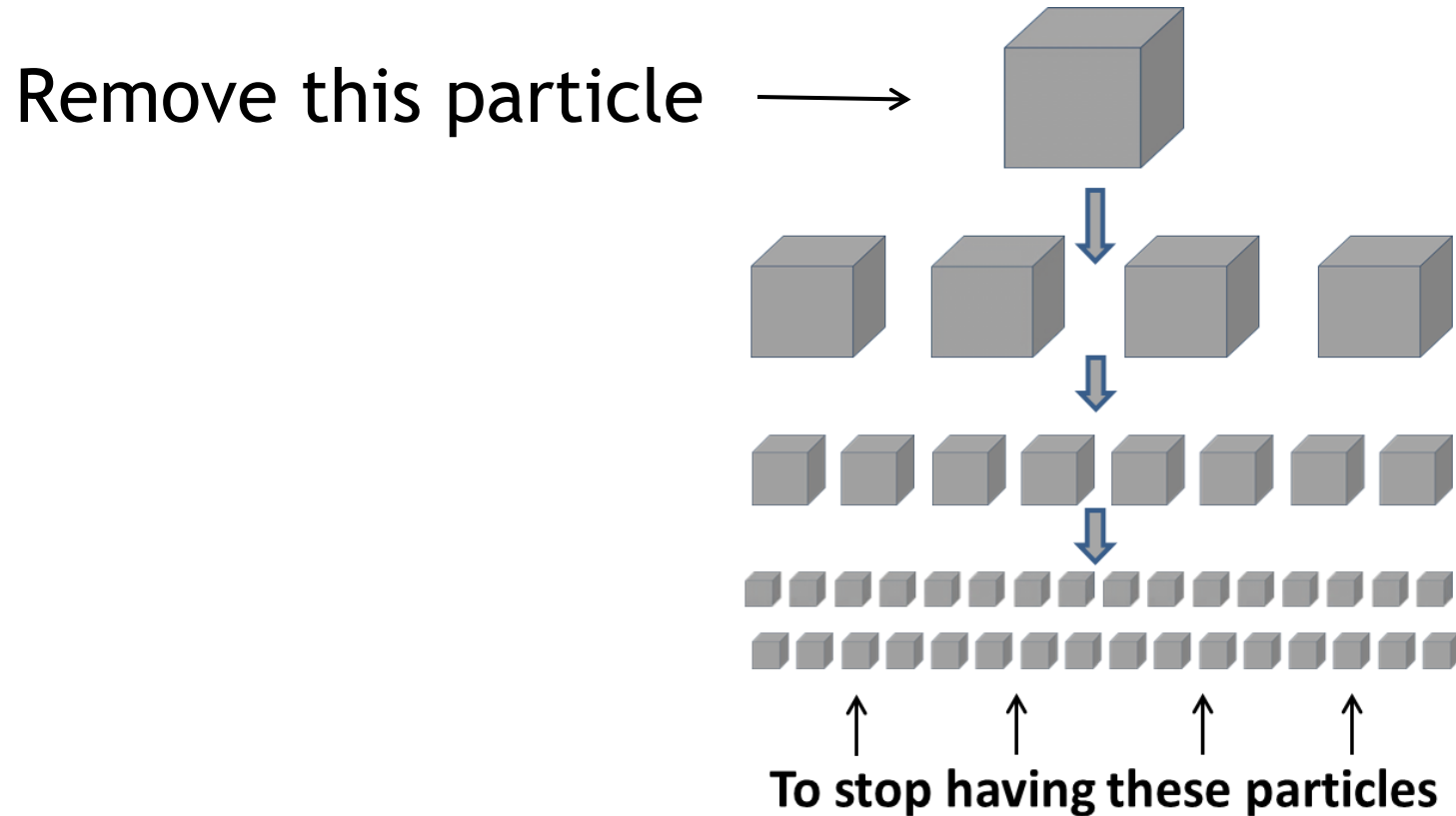
# High Solids as a Contaminant: The Effects

- \* Increased mud weight and mud properties
- \* Swabbing and/or surging
- \* Thicker mud cake; potential tight hole/stuck pipe
- \* Increased dilution rate; increased mud cost
- \* Erosion: due to abrasiveness
- \* Reservoir damage



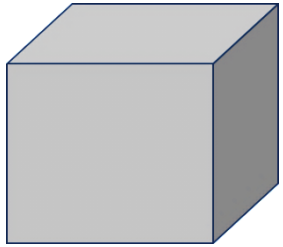
# Solids Degradation

Eliminate solids at first possible opportunity



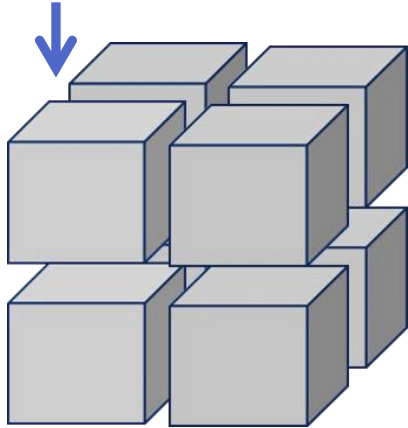
# Particle Degradation

One solid cutting



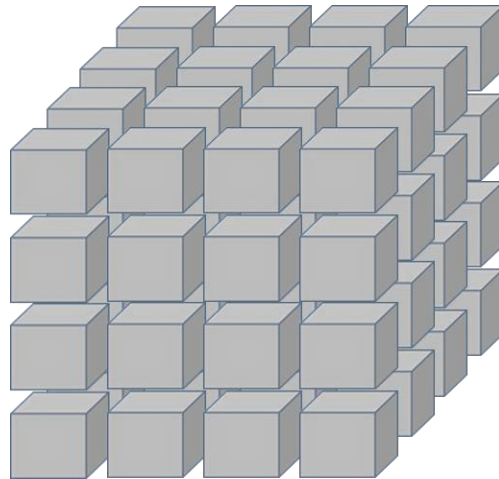
**1in x 1in x 1in**

**Vol. 1in<sup>3</sup> - Area: 6in<sup>2</sup>**



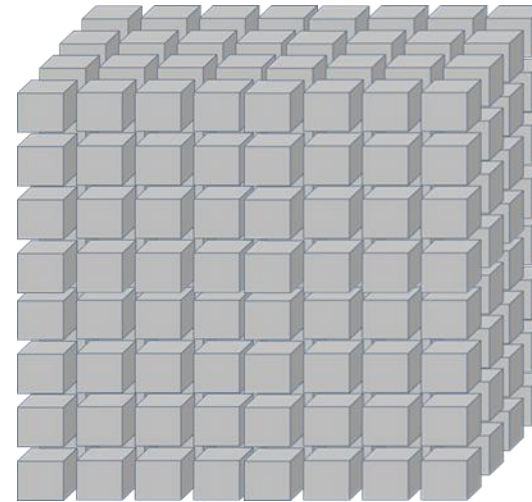
**1/2in x 1/2in x 1/2in**

**Vol. 1in<sup>3</sup> - Area: 12in<sup>2</sup>**



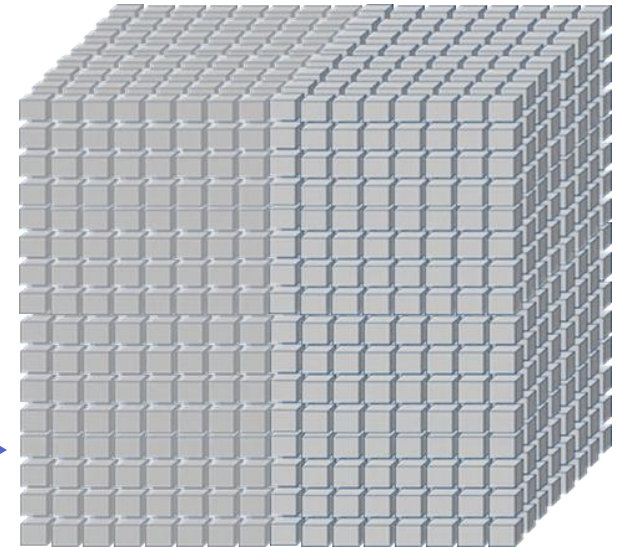
**1/4in x 1/4in x 1/4in**

**Vol. 1in<sup>3</sup> - Area: 24in<sup>2</sup>**



**1/8in x 1/8in x 1/8in**

**Vol. 1in<sup>3</sup> - Area: 48in<sup>2</sup>**



Fine particles (colloids)

**1/16in x 1/16in x 1/16in**

**Vol. 1in<sup>3</sup> - Area: 96in<sup>2</sup>**



# Removing Solids

- \* Mechanical separation: preferable
- \* Dilution
- \* Frequent cleaning of sand traps/settling pits
- \* Dump and Dilute Method



# Shale Shakers

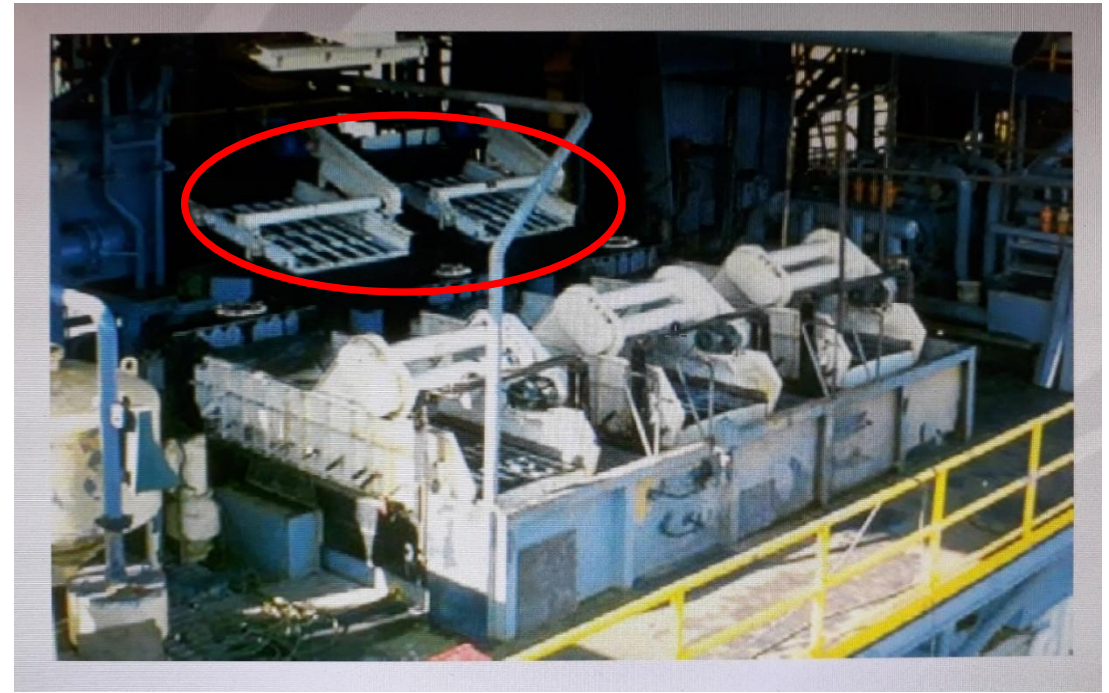
The first mechanical equipment to remove the drilled solids.



# Shale Shakers



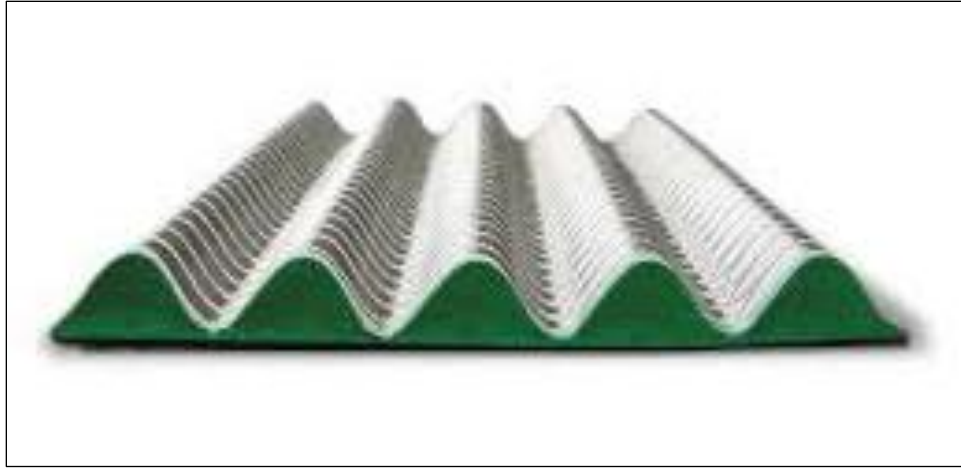
Triple Deck Shale Shaker



Scalping System



# Shaker's Screens Type



Pyramid type screen



Flat type screen

Pyramid type screens provide 125% more effective screening area than a flat type screen.



# Solids Control Equipment: Solids Removal Capability



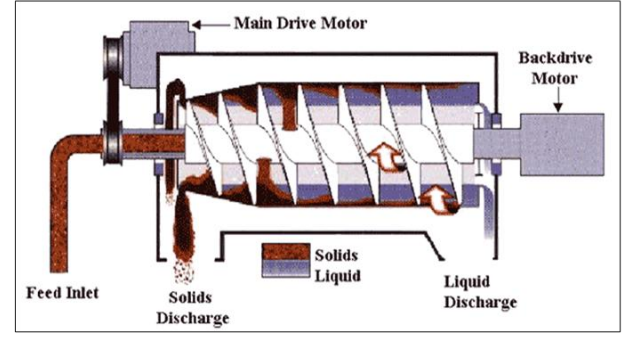
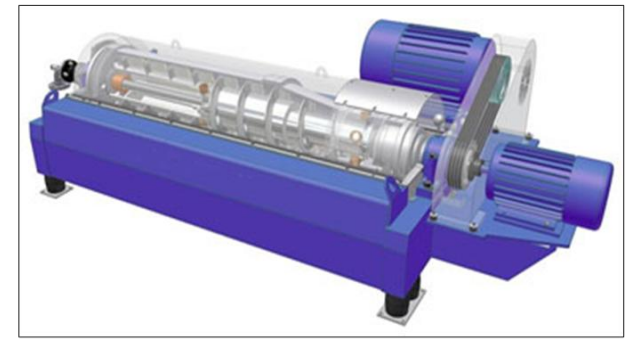
Desander  
Up to 100 $\mu$



Desilter  
Up to 75 $\mu$



Mud Cleaner  
Up to 60 $\mu$



Centrifuge  
Up to 2-10 $\mu$



# Dump and Dilute Method

- \* Implemented if Solids Control Equipment cannot perform effectively.
- \* Mechanism: dump mud volume of highest solids content, replace with newly pre-mixed mud volume
- \* *Advantage*: effective
- \* *Disadvantages*: increased mud cost, environmental concern.



# Conclusions

1. Recognize lithology of well to be drilled to select correct type of a mud system.
2. Drilling and C/WO Fluids should be properly designed and formulated to minimize formation damage.
3. Solids Control Equipment performance should always be at proper running condition.
4. Mud Engineer should be capable to handle the job.

***Protect GOOD reservoir with GOOD MUD***



# THANK YOU



[www.multibinawisesa.com](http://www.multibinawisesa.com)



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Instagram